

Pinpointing Super-Quadratic Quantum Enumeration Speedups

Exact and Certified Evaluation of the Guessing-Moment Exponent under Product-Distribution Advice

Carsten Schubert 
Technische Universität Berlin
Electrical Engineering and Computer
Science
Berlin, Germany
carsten.gm.schubert@tu-berlin.de

Niklas Paskarbeits 
Freie Universität Berlin
Department of Mathematics and
Computer Science
Berlin, Germany

Maximilian J. Kramer 
Freie Universität Berlin
Dahlem Center for Complex
Quantum Systems
Berlin, Germany

Jean-Pierre Seifert 
Technische Universität Berlin
Electrical Engineering and Computer
Science
Berlin, Germany

Marian Margraf 
Freie Universität Berlin
Department of Mathematics and
Computer Science
Berlin, Germany

Abstract

In cryptanalysis and beyond, presumably the most basic problem is search within an unstructured space: Given a finite set of elements K , find a marked element using an efficiently evaluable predicate. Grover’s algorithm [18] gives an optimal quadratic query advantage for black-box search. Cryptanalytic settings, however, often come with additional probabilistic advice over the candidate space. These distributions frequently have product form, e.g. when probabilities for independent key coordinates are derived from side-channel leakage. Classically, simply querying the predicate function in the given likelihood order is optimal to find the correct result as fast as possible in expectation. In the quantum setting, Montanaro [27] showed how to achieve an optimal expected query complexity, beating plain Grover on every non-uniform distribution of the advice (disregarding a constant overhead factor). What has been missing so far is a finite-size method for evaluating the quantum–classical guessing-moment separation induced by a given advice distribution. We provide such a method for product-distribution advice, thereby sharpening the previous entropy-based estimate of Bashiri et al. [7]. Our approach reduces the relevant classical and quantum guessing moments to functionals of the one-dimensional *surprisal* distribution. For product advice, this distribution is obtained by convolving the per-coordinate surprisal laws. When the surprisals lie on a common arithmetic grid—the *commensurate case*—the resulting logarithmic moments, and hence the speedup exponent, can be evaluated as finite sums without discretization error. Exponential tilting makes this computation numerically stable. For general non-commensurate product advice, we discretize the surprisals onto a common grid before running our algorithm and derive an a-posteriori bound on the resulting binning error. We apply the framework to cold-boot leakage on seeds and block-cipher keys, to template-attack posteriors, and to synthetic i.i.d. Bernoulli posteriors whose parameter is chosen to reproduce residual ranks reported for Keccak side-channel attacks on ML-KEM and ML-DSA. The resulting guessing-moment exponents substantially exceed 2 in several skewed-advice settings, reaching up to 3.97 in these synthetic models, and include cases where the previous entropy-based bound did not establish an exponent above 2.

CCS Concepts

• **Theory of computation** → **Quantum query complexity**; *Cryptographic primitives*; • **Security and privacy** → *Cryptanalysis and other attacks*; • **Mathematics of computing** → *Mathematical analysis*.

Keywords

Montanaro’s algorithm, Grover search with advice, quantum key guessing, guessing entropy, Arıkan’s inequality, Rényi entropy, super-quadratic quantum speedup, side-channel analysis

1 Introduction

The most basic task in cryptanalysis is *key guessing*: recover a secret x given only the ability to test candidates.¹ When nothing at all is known about x , the best a quantum attacker can do is Grover search [18]: quadratically fewer guesses than any classical enumeration, and provably no better in this black-box setting. In practice there often is additional information that can be algorithmically exploited by an attacker. Passwords are a familiar example: a capital letter is most likely at the first position, digits and punctuation cluster at the end, and in-between the letters of ordinary language appear at different rates (letter e is more frequent than letter y). Assuming an attacker knows only these positional frequencies, and not how any character influences the next, then the naturally-consistent model is a *product distribution*: one marginal p_i per position $i \in \{1, \dots, m\}$, multiplied together. The resulting distribution P over the key space is the attacker’s *advice*. Similar product shapes arise well beyond this deliberately simplified scenario, further cryptologic examples include independent per-coordinate side-channel leakage [11, 24] or respective secret sampling schemes [9, 29]. Equipped with such advice, the optimal classical attacker iteratively computes an ordering from most to least likely keys and guesses along that list. We define the *rank* $G(x)$ of key x as its position in that list, so reaching x costs $G(x)$ guesses. Thus, the expected cost of the classical attack is the

¹In general, these enumeration and search methods are of course not limited to cryptanalytic settings. But in order to make the formulation more consistent with such typical applications, we stick to their terminology from now on.

average rank $\mathbb{E}_P[G]$, the first *guessing moment*, and it can be far smaller than the full key space. We study the resulting quantum-classical separation in expected query complexity. Specifically, we compute the guessing-moment exponent obtained from the optimal classical moment $\mathbb{E}_P[G]$ and the quantum-relevant moment $\mathbb{E}_P[\sqrt{G}]$, including finite-size and tie effects.

The quantum counterpart of *search with advice* is a more elaborate version of Grover search, due to Montanaro [27]. Instead of searching the whole key space at once, the algorithm walks down the ordered candidate list in segments of geometrically increasing size and runs a Grover search on each one, so that a likely key is found in one of the small early segments and never costs a full search. Reaching a key of rank $G(x)$ this way takes $\Theta(\sqrt{G(x)})$ queries, so the expected query complexity is $\Theta(\mathbb{E}_P[\sqrt{G}])$. Montanaro proved this dependence optimal up to constant factors. Writing $f(\rho) := \ln \mathbb{E}_P[G^\rho]$ for the logarithmic guessing moments, we define the *speedup exponent* $s := f(1)/f(1/2)$. Equivalently, $\mathbb{E}_P[\sqrt{G}] = \mathbb{E}_P[G]^{1/s}$, so the moment governing quantum search is the $1/s$ -th power of the classical expected cost. Montanaro’s bounds transfer this relation to the optimal quantum query complexity up to universal constant factors and an additive overhead; see Section 2. Jensen’s inequality gives $f(\frac{1}{2}) \leq \frac{1}{2}f(1)$ and hence $s \geq 2$. The Grover-like baseline approaches $s = 2$ as the problem size grows. Thus s is a finite-size guessing-moment exponent, while a limiting value bounded away from 2 corresponds to a genuinely *super-quadratic separation*. What has been missing so far is a way to *compute* the advantage exponent s for a given advice distribution, i.e. to tell, at any concrete scheme, how far above 2 that exponent exactly sits.

A first advance in that direction is due to Bashiri et al. [7], who applied Arıkan’s guessing inequality [4] to bound both guessing moments in terms of Rényi entropies of the advice. Their estimate is asymptotically tight, but it bounds s from below only, and does so in the limit of large m . At the finite parameters cryptology cares about, that limit cannot be sharpened: Arıkan’s bounds pin each moment only to within an *additive* slack, and those two slacks, unrelated to one another, leave an interval on the ratio s that does not close. What an attacker cares about, however, is the surplus $s - 2$ over the Grover-like regime, and the unresolved interval can hide a substantial fraction of that. In some parameter settings, the bound from [7] stops cleanly certifying $s > 2$ at all (see Section 7).

In this paper we not only bound s , we also present a method to compute it. Our sole structural assumption is the product form met above, i.e. $P = \prod_{i=1}^m p_i$. For arbitrary product advice, the binned computation is accompanied by a two-sided a-posteriori certificate on its discretization error. The certificate tracks the binning error linearly in the bin width; for bounded alphabets, achieving target accuracy δ therefore costs $\tilde{O}(mC(s)/\delta)$ (Theorem 5.1, Theorem 6.7).² Our main approach is based on one fundamental observation, and a further reduction makes it generally applicable. The observation is that guessing in order of decreasing probability is the same as guessing in order of increasing *surprisal* $S(x) = -\ln P(x)$. Thus, a key’s rank, and with it both guessing moments, is governed by the distribution of this single quantity (Lemma 3.1). Because P

is a product, the surprisal of a key is a *sum* of independent per-coordinate contributions, and the m -dimensional problem collapses to a one-dimensional one. When the per-coordinate surprisals share a common arithmetic grid—we call this the “commensurate” case—that sum is a finite computation and s comes out *exactly* with no discretization error (Theorem 4.1).

Real-world scenarios are usually not commensurate: letter frequencies, for instance, are arbitrary real numbers and share no grid. The subsequent reduction handles every such case in a single step: round the per-coordinate surprisals to a grid of spacing η , forcing commensurability, then run the previously established computation. Under the scaling-law hypotheses of Theorem 6.7 the error this introduces shrinks in proportion to η , so halving the spacing also halves the error. Beyond that scaling law, every run yields an a-posteriori bound on the binning error of the exact estimator, free of unspecified constants or asymptotic ingredients (Proposition 6.9). One mathematical apparatus (exponential tilting), developed once, is used for the exact and general cases and the error analysis alike.

Contributions and roadmap. Section 2 fixes the guessing model, the exponent and the standing assumptions. In Section 3, we reduce both guessing moments to functionals of a single one-dimensional law, turning the speedup exponent from *something to be estimated* into *something computable*. In Section 4, we give an exact method for commensurate advice, evaluating s as a finite sum with no discretization error (Theorem 4.1), and realize it through exponential tilting so that the exponentially small quantities involved remain representable in ordinary double precision (Lemmas 4.2 to 4.4). In Section 5, we lift the commensurability restriction, binning arbitrary advice onto a common grid and bounding both the error incurred and the cost of driving it down (Theorem 5.1), which brings every product distribution within reach. In Section 6, we prove the scaling law governing that error and construct the certificate accompanying each run (Proposition 6.9), so that a reported exponent comes with a non-asymptotic a-posteriori bound on its discretization error rather than an asymptotic promise. Finally, in Section 7, we instantiate the framework on cryptographic advice settings where it is meaningful and read off both the speedup’s size and the earlier bound’s gap. All results are stated in the body, which is self-contained; the appendices carry proofs.

Related work. Montanaro’s routine [27] assumes its advice is already presented in likelihood order, which is not immediate at cryptographic key sizes. Martin et al. [24] gave an algorithm that essentially removed that assumption by returning the r -th key of a weight band efficiently. Later analyses, ours included, assume Martin et al.’s method; their work also first explicitly proposed Montanaro’s quantum search for side-channel attacks. Subsequent variants of quantum search with advice generalize the setting or sharpen overhead constants: He et al. [20] maximize success probability at a fixed query budget, Albrecht and Shen [1] admit bounded-error oracles, and Anderson et al. [3] carry the advice idea to span programs. None of them moves s as we define it, since Montanaro’s two-sided bracket pins it to the problem rather than to any particular routine (see Section 2). Apart from [7], Budroni and Mårtensson [10] are thematically closest to us: they exactly compute classical and quantum query complexity under the stronger assumption that all product marginals are identical. On the defensive side, Fischlin

²Throughout, $\tilde{O}$ suppresses polylogarithmic factors.

et al. [15, 16] show that keys within statistical distance $2^{-\lambda/2}$ of uniform retain at least $\lambda/2$ bits of security. The moments themselves predate the quantum question: Massey [25] showed that Shannon entropy yields no upper bound on $\mathbb{E}_P[G]$, Arıkan [4] supplied the two-sided Rényi bracket, and Rioul [30] develops the full family $\mathbb{E}_P[G^\rho]$ from which $f(1)$ and $f(\frac{1}{2})$ are drawn.

2 Setup and the speedup exponent

The guessing problem. Let $K = \prod_{i=1}^m A_i$ be a product key space with $|A_i| = b_i$ and $N := |K| = \prod_i b_i$, and let the advice be a product distribution $P(x) = \prod_i p_i(x_i)$; write $p_{i,a} := p_i(a)$. The *surprisal* of a key decomposes coordinatewise, $S(x) = -\ln P(x) = \sum_{i=1}^m s_i(x_i)$ with $s_i(a) := -\ln p_{i,a}$. A *guessing order* assigns a distinct position $G(x) \in \{1, \dots, N\}$ to each key, and its cost enters through the moments $\mathbb{E}_P[G^\rho]$, for whose logarithms we write $f(\rho) := \ln \mathbb{E}_P[G^\rho]$. Enumerating by increasing surprisal minimizes these moments simultaneously for every $\rho > 0$, and every such order attains the same value (Lemma 3.1). Keys of equal surprisal may therefore be enumerated in any order, and no tie convention enters the moments. We write G for any optimal order and reserve $G^*(x) := \#\{x' \in K : S(x') \leq S(x)\}$ for the *block-maximal* rank, the largest rank in x 's equal-surprisal block. Thus $G(x) \leq G^*(x)$; equality holds for the key placed last within its tie block, and for every key on that level iff the block is a singleton. Logarithms are natural throughout — surprisals, entropies, the bin width η and all saddle quantities are in nats — while s , being a ratio of logarithms, is base-free.

Cost model and the exponent. By W_C and W_Q we denote the optimal classical and quantum query complexities of guessing under advice P . Classically, $W_C = \mathbb{E}_P[G]$ exactly. Montanaro's bounds [27] show that, up to universal constant factors, the quantum complexity is governed by the square-root guessing moment, $W_Q = \Theta(\mathbb{E}_P[\sqrt{G}])$; the algorithm attaining the upper bound additionally incurs a lower-order additive overhead. Consequently, $\ln W_C = f(1)$ and $\ln W_Q = f(\frac{1}{2}) + O(1)$. We therefore define the information-theoretic *speedup exponent* $s := f(1)/f(\frac{1}{2})$. The corresponding operational logarithmic exponent satisfies $\frac{\ln W_C}{\ln W_Q} = s(1 + O(1/f(\frac{1}{2})))$, and hence approaches s whenever $f(\frac{1}{2}) \rightarrow \infty$; for families with $f(\frac{1}{2}) = \Theta(m)$ the relative discrepancy is $O(1/m)$. All subsequent statements concern this guessing-moment exponent s . Disregarding the operational overhead like that, we note that s naturally already exceeds 2 at every *finite* N even for uniform advice, but in contrast to non-uniform advice it *asymptotically* approaches 2 there as $N \rightarrow \infty$.

Information-theoretic quantities. Two distributions on K are in play: a key's rank counts how many keys are at least as likely, which therefore refers to the uniform distribution U on K , while the guessing moments average under the advice P . To address this, we use the term *law* interchangeably with the distribution. For $\alpha \in (0, 1)$, the Rényi entropy $H_\alpha(P) = \frac{1}{1-\alpha} \ln \sum_x P(x)^\alpha$ is additive for product laws, it has the Shannon entropy $H_1(P) := -\sum_x P(x) \ln P(x)$ as its continuous extension. Write $\bar{H}_\alpha := \frac{1}{m} \sum_i H_\alpha(p_i)$ and $H := \mathbb{E}_P[S] = H_1(P) = m\bar{H}_1$.

Under U set $\mu_U = \mathbb{E}_U[S]$, $\sigma_U^2 = \text{Var}_U[S]$ and $\Gamma := \mu_U - H \geq 0$, where $\Gamma = D(P\|U) + D(U\|P)$ is the Jeffreys divergence; it is $\Theta(m)$ for fixed non-uniform per-coordinate families. Arıkan's guessing

Table 1: Recurring notation.

symbol	meaning
K, A_i, b_i, m, N	key space $K = \prod_{i=1}^m A_i$; alphabets, $ A_i = b_i$; $N = K $ keys; input size
$B_{\text{in}} = \sum_i b_i$	product posterior (advice), marginals p_i ; uniform law on K
$P = \prod_i p_i, U$	product posterior (advice), marginals p_i ; uniform law on K
$s_i(a) = -\ln p_{i,a}$	per-coordinate surprisal
$S = \sum_i s_i$	total surprisal
$\pi, v; F_S$	laws of S under P, U ; uniform CDF $F_S(t) = \mathbb{P}_U[S \leq t]$
$G; G^* = N \cdot F_S(S)$	optimal guessing order; block-maximal rank (Lemma 3.1)
$\mathcal{A}_\rho(t); \bar{\mathcal{A}}_\rho$	level mean power rank $n(t)^{-1} \sum_{r \in \mathcal{A}_\rho} r^\rho$ (Lemma 3.1); binned analogue
$f(\rho), s$	log guessing moment $\ln \mathbb{E}_P[G^\rho]$; exponent $s = f(1)/f(\frac{1}{2})$
$H_\alpha, \bar{H}_\alpha$	Rényi entropy and its per-coordinate mean $\frac{1}{m} \sum_i H_\alpha(p_i)$
H	Shannon entropy $H = \mathbb{E}_P[S] = m\bar{H}_1$
$\mu_U, \sigma_U^2; \Gamma$	mean, variance of S under U ; gap $\Gamma = \mu_U - H$ (Jeffreys divergence)
$\Lambda_U(\tau)$	uniform CGF $\ln \mathbb{E}_U[e^{\tau S}]$; $\Lambda'_U(0) = \mu_U, \Lambda''_U(0) = \sigma_U^2$
$U_\tau; \sigma_\tau^2$	tilted law $U_\tau^{(t)} \propto e^{\tau s_i}, U_{-1} = P; \sigma_\tau^2 = \Lambda''_U(\tau)$
$\tau_t, \sigma_t; I_U(t)$	saddle $\Lambda'_U(\tau_t) = t$, scale $\sigma_t := \sigma_{\tau_t}$; rate function $\sup_\tau (\tau t - \Lambda_U(\tau))$
$\theta(t), \theta^*$	rank-growth rate $ \tau_t $; edge value $\theta^* = \tau_H = 1$ (Corollary 6.5)
$\phi, \mu(\phi)$	engine tilt and tilted grid law (Section 4.2)
$\eta, h; R; e_0$	bin width / lattice span; surprisal range; scaling-law regime constant
$\Lambda_\eta; q_\tau$	lattice-span exponent (10); quartic correction (Lemma 6.3)
$\tilde{s}_i, \tilde{S}, \tilde{\pi}, \tilde{v}, \tilde{F}, \tilde{G}$	binned analogues: strictly upward rounding to $\eta\mathbb{Z}$, $0 < \tilde{S} - S \leq m\eta$
$\tilde{f}_\eta, \tilde{s}_\eta; \tilde{S}$	binned moments, exponent; floating-point output (Lemma 4.4)
$\Delta(x)$	binning window log-ratio, floored at rank 1 (Lemma 6.1)
$C(s), e_m$	scaling-law constant $(2+s)/\bar{H}_{2/3}$; finite-size correction of Theorem 6.7
$\text{Var}_1; \sigma_H^2$	mean per-coordinate surprisal variance under P ; $\sigma_H^2 = m\text{Var}_1$
$\varepsilon_\tau, \beta_\tau$	local-CLT error, summed tilted third moment (Lemma 6.3)
$\delta, B; u_m, \text{poly}$	target precision; a-posteriori error bound (Proposition 6.9); unit roundoff; polynomial in $m, 1/\eta$

inequality [4] gives

$$f(\rho) = \rho \sum_{i=1}^m H_{1/(1+\rho)}(p_i) + O(\log(1 + \ln N)). \quad (1)$$

For uniformly bounded alphabets this yields $f(\rho) = \rho m \bar{H}_{1/(1+\rho)}(1 + O(\log m/m))$ whenever the per-coordinate entropy stays bounded away from zero, and hence $s = 2\bar{H}_{1/2}/\bar{H}_{2/3} (1 + o(1))$. This recovers the asymptotic entropy scale underlying [7]; see Appendix A.

Standing assumptions. We assume throughout:

- (i) P is a product distribution, which is what makes the surprisal additive and the whole analysis one-dimensional; advice with genuine dependence between coordinates is out of scope, though richer models often fit by grouping positions into coarser coordinates and taking their product.
- (ii) $p_{i,a} > 0$ for every i, a is a convention rather than a restriction: symbols of zero probability can simply be removed from A_i .
- (iii) Complexity is measured in the input size $B_{\text{in}} := \sum_i b_i$ and, where discretization enters, the level count; no uniform bound on the alphabet sizes is assumed.
- (iv) For the saddle-point analysis we assume non-degeneracy, $\text{Var}_U(S) > 0$, equivalently that P is not uniform. The uniform case is excluded only from that analysis and can be evaluated directly.

Notation. Table 1 collects the recurring symbols.

3 The rank–CDF reduction

The guessing moments average over all N keys, where N can be exponential in m , so direct enumeration is infeasible. The next lemma removes the key space from the picture, replacing it with two one-dimensional distributions of the surprisal: how many keys

sit at each surprisal level, and how likely the secret is to sit there. Because P is a product, each is an m -fold convolution of m small per-coordinate distributions. Every subsequent algorithm in this paper computes those convolutions.

The approach works because the surprisal of a key, while it does not fix that key's rank, confines it narrowly: keys of equal surprisal occupy consecutive ranks, and the moment averages over those ranks exactly. Individual ranks therefore depend on how ties are broken, but the moments do not.

Fix a surprisal level t . Counting keys is an operation under the uniform distribution U , so we write $F_S(t) := \mathbb{P}_U[S \leq t]$, $F_S(t^-) := \mathbb{P}_U[S < t]$ and $v(t) := \mathbb{P}_U[S = t]$ for the cumulative distribution function of S under U , its left limit, and its mass at t . The first of these is what a rank measures: $N \cdot F_S(t)$ is exactly the number of keys at least as likely as one of surprisal t , and $n(t) := N \cdot v(t)$ is the number of keys at that level. Accordingly we set

$$J_t := \{NF_S(t^-) + 1, \dots, NF_S(t)\}, \text{ and } \mathcal{A}_\rho(t) := \frac{1}{n(t)} \sum_{r \in J_t} r^\rho,$$

an integer interval containing exactly $n(t)$ ranks and the mean ρ -th power of a rank across it. The advice enters at one point only, through $\pi(t) := \mathbb{P}_P[S = t]$, the probability that the secret has surprisal t ; since every key at level t has probability e^{-t} , these are related by $\pi(t) = n(t)e^{-t}$.

LEMMA 3.1 (RANK-CDF REDUCTION).

- (i) Among guessing orders, each moment $\mathbb{E}_P[G^\rho]$ is minimized exactly by the likelihood-ordered ones (for $\rho > 0$), and the value is invariant under permutations within equal-surprisal blocks (proof in Appendix E).
- (ii) Every optimal G satisfies $N \cdot F_S(S(x)^-) < G(x) \leq N \cdot F_S(S(x)) = G^*(x)$; that is, the ranks on level t fill J_t .
- (iii) For every $\rho > 0$, $\mathbb{E}_P[G^\rho] = \sum_t \pi(t) \mathcal{A}_\rho(t)$ and therefore $f(\rho) = \ln \sum_t \pi(t) \mathcal{A}_\rho(t)$.

PROOF. (ii) Since $t \mapsto e^{-t}$ is strictly decreasing, a likelihood order ranks every key of surprisal $< t$ before, and every key of surprisal $> t$ after, the $n(t)$ keys at level t ; those keys therefore occupy exactly the ranks of J_t . (iii) On $\{S = t\}$ every key has probability e^{-t} , so $\sum_{x: S(x)=t} P(x) G(x)^\rho = e^{-t} \sum_{r \in J_t} r^\rho = \pi(t) \mathcal{A}_\rho(t)$; sum over levels. By (i) the value does not depend on which optimal G was chosen. $\square$

Item (iii) is the identity every later algorithm evaluates. What remains is to compute the two surprisal distributions it involves: without discretization error when the per-coordinate surprisals are commensurate (Section 4), and otherwise through the binned approximation and its a-posteriori error bound (Sections 5 and 6).

4 Exact evaluation and the stable engine

This section describes the main computation engine. For commensurate marginals, s is given *exactly* by a finite sum, with no discretization error (Section 4.1); Section 4.2 gives its numerically stable floating-point realization for typical m . Throughout this and the next section, poly denotes a fixed polynomial in m , $1/\eta$, and h may be read for η in the commensurate case. Cost statements are in *word* operations at fixed precision, which Lemma 4.4 justifies by showing double precision suffices for the targets of interest;

under the interval-arithmetic route the working precision becomes a parameter and the bit complexity scales with it.

4.1 The exact evaluation

Say the marginals are *commensurate* if all $\ln p_{i,a}$ lie on a common arithmetic grid, $s_i(a) \in h\mathbb{Z}$ for some span $h > 0$. Then S takes values in $h\mathbb{Z}$, and since it ranges over an interval of width $R := \sum_i (\max_a s_i(a) - \min_a s_i(a))$, the *surprisal range*, its support has $O(R/h)$ points (polynomial when the range is $O(m)$, as under bounded per-coordinate leakage). Dyadic models — and more generally all models whose probabilities are powers of a common base, $p_{i,a} = q^{-k_{i,a}}$, so that their surprisals are specified in fixed units — are commensurate by construction.

On such a grid Lemma 3.1 (iii) becomes a finite computation: the level sum runs over the $O(R/h)$ grid points, and $\mathcal{A}_\rho(t)$ depends on the level only through the pair $(F_S(t^-), F_S(t))$. The two histograms promised in Section 3 are therefore convolutions of explicit per-coordinate laws: the *mass* laws $a \mapsto p_{i,a}$ convolve to π , the *count* laws $a \mapsto 1/b_i$ convolve to what we call v .

THEOREM 4.1 (EXACT EVALUATION). *In the commensurate case*

$$f(\rho) = \rho \ln N + \ln \sum_t \pi(t) \frac{\mathcal{A}_\rho(t)}{N^\rho}$$

is a finite sum over the grid — the factor N^ρ split off so that every summand lies in $[0, 1]$ — and $s = f(1)/f(\frac{1}{2})$ is computed exactly in $\tilde{O}(R/h + B_{\text{in}})$ time, where $B_{\text{in}} = \sum_i b_i$ is the input size; for extensive range $R = O(m)$ this is $\tilde{O}(m/h)$. Moreover, at every level,

$$\frac{1}{1+\rho} \leq \frac{\mathcal{A}_\rho(t)}{(N \cdot F_S(t))^\rho} \leq 1, \quad (2)$$

with equality on the right iff the level is a singleton ($n(t) = 1$): the block-maximal surrogate $(N \cdot F_S(t))^\rho$ is exact precisely when no two keys share a surprisal.

PROOF. The display is the level sum of Lemma 3.1 (iii) under the scale split $r^\rho = N^\rho (r/N)^\rho$; the bracket (2) and the certified $O(1)$ -per-level evaluation of $\mathcal{A}_\rho$ from $(F_S(t^-), F_S(t))$ are proved in Appendix E. It remains to produce π and v and to cumulate v into F_S . A grid law is a coefficient vector, convolving two of them is polynomial multiplication, and the FFT performs it in $O(n \log n)$ rather than $O(n^2)$ on supports of n points. Folding the m per-coordinate laws in one at a time would re-traverse the growing support $\Theta(m)$ times so instead we convolve them along a *balanced binary merge tree*, as in mergesort — coordinates in pairs, then the results in pairs, for $\lceil \log_2 m \rceil$ rounds — in exact arithmetic here (the floating-point realization is Section 4.2), sizing each FFT to the actual support of its output. Supports of disjoint blocks add, so the nodes of any one round jointly hold $O(R/h + m)$ grid points and cost $O((R/h + m) \log(R/h + m))$; over the rounds this is $\tilde{O}(R/h + m)$ as each grid cell is touched at $O(\log m)$ nodes. The cumulative sum and the two moment sums are linear. $\square$

The two convolutions are in fact one: On the level set $\{S = t\}$ every key has probability e^{-t} , so mass is count times weight:

$$\pi(t) = n(t)e^{-t} = N v(t) e^{-t} \quad (\text{exactly}). \quad (3)$$

Summing (3) over levels $u \leq t$ and using $\sum_{u \leq t} \pi(u) \leq 1$ gives an exact Chernoff-type bound we will reuse:

$$N \cdot F_S(t) = \sum_{u \leq t} e^u \pi(u) \leq e^t, \quad \text{i.e.} \quad G^* \leq e^S \text{ pointwise; (4)}$$

a key's rank never exceeds one over its probability, with near-equality at typical keys, where $S \approx H$. The exact engine therefore needs only v : cumulate it into F_S , and recover π from (3). This collapse fails under binning, since keys rounded into one cell no longer share a weight, which is why the following Section 5 keeps two *chains* — our term for one histogram carried through the tilt-merge-cumulate pipeline.

The finite-rank exact engine. Commensurability is the case $d = 1$ of a broader hypothesis under which the level count stays finite: Say the marginals have *rank* d if there are reals $\beta_1, \dots, \beta_d$, linearly independent over $\mathbb{Q}$, constants c_i and integers $c_{i,a,j}$ with $s_i(a) = c_i + \sum_{j=1}^d c_{i,a,j} \beta_j$ for every i, a — the c_i shift every key's surprisal by the same $\sum_i c_i$, changing neither order nor moments. The rank and the β_j belong to the *model specification* and are not recovered from the numbers — detecting $\mathbb{Q}$ -linear relations among given reals is a different and much harder problem — but under structured advice they may come for free: for a repeated k -symbol law the logarithms $-\ln q_1, \dots, -\ln q_k$ span a $\mathbb{Q}$ -vector space of dimension $d \leq k$. A basis of that span, with denominators cleared, puts the surprisals in the form above. Bounded d is what divides the paper's two routes: ternary secrets sit at small d , while heterogeneous leakage posteriors generically have rank $\Theta(m)$, so the latter regime belongs to the binned engine (Section 5).

Bounded rank keeps the level count polynomial. Collect the integer vectors $W := \{\sum_i c_{i,x_i} : x \in K\} \subseteq \mathbb{Z}^d$. The map $w \mapsto \langle w, \beta \rangle$ is injective on $\mathbb{Z}^d$ — a coincidence of values would be a nontrivial rational relation among the β_j — so the levels of S correspond exactly to the elements of W , and these lie in a box whose j -th side is the total spread of $c_{\cdot,j}$ over the coordinates. When those spreads are extensive, the rank- d analogue of an extensive range, the box holds $O(m^d)$ points, reducing to $O(R/h)$ at $d = 1$ (Appendix E). The engine then carries over with $\mathbb{Z}^d$ in place of $h\mathbb{Z}$: per-coordinate laws become measures on $\mathbb{Z}^d$ supported on b_i points, and the balanced merge uses d -dimensional FFTs sized to the actual support boxes, $\tilde{O}(m^d)$ in total. Algorithm 1 below is stated for $d = 1$ for simplicity. For higher ranks, read $\mathbb{Z}^d$ for $h\mathbb{Z}$ throughout it. Additionally, one step is genuinely new. At $d = 1$ the grid $h\mathbb{Z}$ arrives already ordered, so cumulating v into F_S is immediate; for $d \geq 2$ the surprisal order is induced by the linear functional $\langle \cdot, \beta \rangle$, and the $O(m^d)$ levels must be sorted by $\langle w, \beta \rangle$ before they can be cumulated. The finish is untouched: cumulate, recover π from (3), apply Theorem 4.1.

That sort is where exactness meets its limit. Ordering requires deciding the sign of $\langle w - w', \beta \rangle$, a nonzero real, which adaptive-precision interval arithmetic settles in finitely many refinements whenever the β_j are available to arbitrary precision, as logarithms of rationals are — but the precision required is not bounded a priori. The fallback is a device this paper already carries: levels that working precision cannot separate are merged into one block and averaged by $\mathcal{A}_\rho$. Merging replaces the favorable pairing of largest masses with smallest ranks by a neutral one, so by Lemma 3.1(i) it can only *overstate* the moment; the merged value is a certified upper

bound. The matching lower bound is elementary: a merged block occupies a *consecutive* rank interval $\{a+1, \dots, a+n\}$ of mass Π , so its true contribution is at least $\Pi(a+1)^\rho$ against a merged value of at most $\Pi(a+n)^\rho$. Summing these widths over the merged blocks, the engine returns a certified enclosure in place of an exact value — degrading into the same two-sided mode as the binned route, by the same machinery. Where the level count stays polynomial this engine is cheap, and it is precisely there that the binned certificate's precision later floors (Remark 6.10): the two are complementary rather than competing, with a band at slowly growing d in which both run and neither dominates.

4.2 Stable evaluation

At typical m the dominant summands $\pi(t)F_S(t)^\rho$ are $e^{-\Theta(m)}$ and underflow in double precision. But their smallness is benign: at every level it factors into an exponential scale that is available in closed form, times a shape of moderate dynamic range (see below). The scheme below maintains this factorization through its whole computation: floating point only ever touches the shape, while the scale travels symbolically in the log domain; in effect, a mantissa-exponent split for the entire pipeline. We state it first and then justify its steps in turn. Three lemmas make it rigorous, of which two are proved in Appendix E.

Algorithm 1 — Stable exact evaluation (commensurate P)

Input: commensurate marginals on $h\mathbb{Z}$.

Output: speedup s , accurate to floating point (Lemma 4.4).

- (1) Form the per-coordinate count histograms v_i on $h\mathbb{Z}$.
 - (2) **For** each $\rho \in \{1, \frac{1}{2}\}$ **do**
 - (a) Tilt every v_i by a common ϕ , chosen so the merged law is centered on the levels dominating $f(\rho)$; by Lemma 4.2 this is exact.
 - (b) FFT-merge the tilted $v_i^{(\phi)}$ up a balanced tree (support-aware, truncated; Lemma 4.3).
 - (c) In the log domain, recover $\ln F_S$ by the left-to-right prefix LSE of Eq. (5) and $\ln \pi$ from Eq. (3), so the $e^{-\Theta(m)}$ magnitudes never leave the analytic factor.
 - (d) Assemble $f(\rho) = \text{LSE}_t [\ln \pi(t) + \ln(N^{-\rho} \mathcal{A}_\rho(t))] + \rho \ln N$, each $N^{-\rho} \mathcal{A}_\rho(t)$ evaluated in $O(1)$ from the prefix pair $(F_S(t^-), F_S(t))$ to certified precision (shown by Appendix E).
 - (3) Output $s = f(1)/f(\frac{1}{2})$.
-

The tilt (step a). The device effecting the split is the exponential *tilt*: for a grid law μ with cumulant generating function (CGF) $\Lambda_\mu(\phi) = \ln \sum_w e^{\phi w} \mu(w)$, set $\mu^{(\phi)}(w) := e^{\phi w - \Lambda_\mu(\phi)} \mu(w)$ — an exponential ramp across the levels, re-normalized into a probability law with mean $\Lambda'_\mu(\phi)$. Choosing ϕ so that this mean lands on the levels dominating the moment sums (the *saddle* levels) makes the ramp cancel the $e^{-\Theta(m)}$ exactly there: the entries the answer depends on become well-scaled entries of $\mu^{(\phi)}$, and what was divided out is known exactly, namely $e^{\phi v - \Lambda_\mu(\phi)}$. The engine tilts the count law v , whose CGF is the uniform CGF $\Lambda_U(\tau) =$

$\ln \mathbb{E}_U[e^{\tau S}] = \sum_i \lambda_i(\tau)$, strictly convex, with per-coordinate summands $\lambda_i(\tau) = \ln \frac{1}{b_i} \sum_a e^{\tau s_i(a)}$; on this chain the generic tilt parameter ϕ is precisely τ , and $U_{-1} = P$ since $U_{-1}^{(i)}(a) \propto e^{\tau s_i(a)} = p_{i,a}^{-\tau}$.

The centring is a numerical device, not a correctness ingredient: any ϕ that keeps the contributing entries within the truncation window of Lemma 4.3 is admissible (a few Newton steps on the strictly increasing Λ'_U if one wants the exact saddle). What the step does require is that tilting survive the merge.

LEMMA 4.2 (TILT COMMUTES WITH CONVOLUTION). *If $\mu = \mu_1 * \mu_2$ with CGFs λ_1, λ_2 , then $\mu^{(\phi)} = \mu_1^{(\phi)} * \mu_2^{(\phi)}$ and the CGFs add, $\Lambda_\mu = \lambda_1 + \lambda_2$; hence $(*_i \mu_i)^{(\phi)} = *_i \mu_i^{(\phi)}$ with CGF $\sum_i \lambda_i$.*

PROOF. For $a + b = w$, $e^{\phi a} e^{\phi b} = e^{\phi w}$, so $(\mu_1^{(\phi)} * \mu_2^{(\phi)})(w) = e^{\phi w - \lambda_1 - \lambda_2} \sum_{a+b=w} \mu_1(a) \mu_2(b) = \mu^{(\phi)}(w)$; induct on m . $\square$

Convolving the tilted per-coordinate laws therefore returns exactly the tilt of the true law: tilting carries no bias, so after the merge we hold $\mu^{(\phi)}$ for the true μ and may simply solve the definition of the tilt for μ , level by level.

The merge (step b). We call a chain $\frac{1}{2}$ -decaying if within every coordinate the tilted atom weights fall off exponentially in the surprisal, at a rate of at least $\frac{1}{2}$ per unit above the per-coordinate minimum. Both chains are $\frac{1}{2}$ -decaying at every tilt either engine uses (Algorithms 1 and 2), uniformly in the surprisal range — the negative effective tilt is what makes long tails cheap.

LEMMA 4.3 (MERGE COST). (i) *The support-aware balanced merge of the m per-coordinate laws, plus a cumulative sum, costs $\tilde{O}(R/\eta + B_{\text{in}})$ time and $\Theta(R/\eta + m)$ space.* (ii) *For $\frac{1}{2}$ -decaying chains, each partial convolution over a block I of coordinates carries all but a poly^{-1} fraction of its mass on $\tilde{O}(\sqrt{B_I}/\eta)$ grid points around its analytically known mean, where $B_I := \sum_{i \in I} b_i$ is the block input size; truncating every merge node to this window perturbs the output by poly^{-1} in ℓ_1 (absorbed into Lemma 4.4; see Appendix E) and the cost drops to $\tilde{O}((\sqrt{m B_{\text{in}}} + m)/\eta)$ time — $\tilde{O}(m/\eta)$ for bounded alphabets — and $\tilde{O}(\sqrt{B_{\text{in}}}/\eta)$ space, independently of R .*

Thus, the truncated engine costs $\tilde{O}((\sqrt{m B_{\text{in}}} + m)/\eta)$, and $\tilde{O}(m/\eta)$ for uniformly bounded alphabets, independently of the surprisal range.

Log-domain recovery (step c). Inverting the tilt is the first line of (5) below, and its two ingredients are precisely the split announced above: an analytic factor $-\phi v + \Lambda(\phi)$, of size $\Theta(m)$ but known in closed form (Lemma 4.2 assembles $\Lambda = \sum_i \lambda_i$ from the marginals), and a computable $\ln \mu^{(\phi)}(w)$, of moderate size by the centring. Applying this to v and summing over the levels $v \leq t$ gives the second line. We do this in the log domain and hence use “log-sum-exps” (LSE):

$$\ln \mu(w) = \underbrace{-\phi w + \Lambda(\phi)}_{\text{analytic, } \Theta(m)} + \ln \mu^{(\phi)}(w), \quad (5)$$

$$\ln F_S(t) = \Lambda_U(\phi) + \text{LSE}_{w \leq t} \left[-\phi w + \ln \nu^{(\phi)}(w) \right],$$

where $\text{LSE}_w[x_w] := \ln \sum_w e^{x_w}$. Log-sum-exps appear at two points, and are evaluated differently because they are needed differently. The assembly of $f(\rho)$ in step d requires a single LSE over all levels,

computed once. There the batch form applies: shift by the maximum, $\text{LSE}_w[x_w] = \max_w x_w + \ln \sum_w e^{x_w - \max_u x_u}$, so that every exponent is ≤ 0 . However, the cumulation into $\ln F_S$ in this step is another problem: it is needed at every level t , and treating each as a separate batch LSE would cost quadratic time in the number of levels. Instead a single left-to-right sweep maintains the running prefix $L_t := \text{LSE}_{w \leq t}[x_w]$, folding in one level at a time by the identity

$$L_t = \max(L_{t-1}, x_t) + \ln \left(1 + e^{-|L_{t-1} - x_t|} \right).$$

The larger of the two numbers is carried outside the logarithm unchanged; the smaller enters only through the correction term, which lies in $(0, \ln 2]$. The exponent is never positive, so nothing overflows and each level costs $O(1)$.

Assembly and stability (step d). At the assembly, we have:

LEMMA 4.4 (FLOATING-POINT STABILITY). *Algorithm 1 returns $\hat{s}$ with*

$$|\hat{s} - s_\eta| = O \left(\frac{u_m \text{poly}(m, 1/\eta) (1 + s_\eta)}{f_\eta(\frac{1}{2})} \right),$$

where u_m is the floating-point unit roundoff; s_η and $f_\eta(\frac{1}{2})$ are the exact values of the estimator being run ($s_\eta = s$ and $f_\eta = f$ in the commensurate case), provided the roundoff in $f_\eta(\frac{1}{2})$ is smaller than a constant fraction of $f_\eta(\frac{1}{2})$. For families with $f_\eta(\frac{1}{2}) = \Theta(m)$ and $s_\eta = O(1)$, this simplifies to $|\hat{s} - s_\eta| = O(u_m \text{poly}(m, 1/\eta)/m)$.

For families with $f_\eta(\frac{1}{2}) = \Theta(m)$ and $s_\eta = O(1)$, this is negligible against target precisions $\delta \gg u_m \cdot \text{poly}/m$; in IEEE double, $u_m \approx 1.1 \times 10^{-16}$. The polynomial in Lemma 4.4 is the only constant in the pipeline whose growth in m and $1/\eta$ is unspecified; those of Section 6 are absolute. It governs floating-point roundoff, not the binning error that Proposition 6.9 later certifies, and it is removable at the implementation level: the entire assembly is a composition of FFTs, sums and 1-Lipschitz LSEs, so running it in directed-rounding interval arithmetic returns a rigorous enclosure end-to-end at a constant-factor cost — the certificate of Proposition 6.9 then certifies its own arithmetic. We state this as an implementation route, not as a property of the reference implementation.

The full proofs of both lemmas are deferred to Appendix E, so here we focus on delineating their underlying mechanism: after tilting, the grid entries the output depends on sit within a poly factor of the norms of the vectors the FFTs handle. Thus, Higham’s norm-wise bound, which is $O(u_m \log L)$ for a length- L transform [21] composed up a $\lceil \log_2 m \rceil$ -deep tree, leaves them with $u_m \cdot \text{poly}$ relative precision. The $e^{-\Theta(m)}$ saddle magnitudes never materialize as floats, living instead in the analytic factor of (5), so nothing underflows; the log-domain assembly is a composition of 1-Lipschitz LSEs; and the final ratio contributes the conditioning factor $(1 + s_\eta)/f_\eta(\frac{1}{2})$; when $f_\eta(\frac{1}{2}) = \Theta(m)$ and $s_\eta = O(1)$, this supplies the stated $1/m$ factor.

5 Binning: the general case

A measured posterior is generically *not* commensurate: each coordinate has only b_i surprisal values, but the total S takes up to $\prod_i b_i$ distinct values in a bounded range, so direct enumeration is infeasible. The reduction is essentially one step: snap the per-coordinate

surprisals to a common grid $\eta\mathbb{Z}$, forcing commensurability, and run the engine (see Algorithm 2).

Bin the surprisals, not the probabilities. Round each per-coordinate surprisal *strictly* up onto $\eta\mathbb{Z}$, $\tilde{s}_i(a) := \eta(\lfloor s_i(a)/\eta \rfloor + 1) \in (s_i(a), s_i(a) + \eta]$, so that $0 < \tilde{S}(x) - S(x) \leq m\eta$ for every key.³ It is essential to bin the surprisals rather than the probabilities: rounding probabilities would place the surprisals on an irregular set whose m -fold sums have super-polynomially many distinct values and do not collapse. Snapping surprisals to $\eta\mathbb{Z}$ puts every sum on $\eta\mathbb{Z}$, with support $O(R/\eta)$.

Two chains and the estimator. Because (3) fails after binning, the engine runs on two per-coordinate histograms: a P -mass histogram π_i (bin k accumulates $\sum_{a:\tilde{s}_i(a)=k\eta} p_{i,a}$, keeping the masses exact) and a U -count histogram v_i . Convolving gives $\tilde{\pi}$ (law of $\tilde{S}$ under the true weights P) and $\tilde{v}$ (count under U), with $\tilde{F}$ its cumulative sum. The estimator is the finite sum of Theorem 4.1 with $(\tilde{\pi}, \tilde{F})$ in place of (π, F_S) : with $\tilde{\mathcal{A}}_\rho(t)$ the level mean power rank of the binned law (built from $\tilde{F}(t^-)$, $\tilde{F}(t)$ exactly as in Lemma 3.1),

$$f_\eta(\rho) := \rho \ln N + \ln \sum_t \tilde{\pi}(t) \frac{\tilde{\mathcal{A}}_\rho(t)}{N\rho} = \ln \mathbb{E}_P[\tilde{\mathcal{A}}_\rho(\tilde{S})], \quad (6)$$

$$\tilde{G}(x) := N \cdot \tilde{F}(\tilde{S}(x)), \quad s_\eta := \frac{f_\eta(1)}{f_\eta(\frac{1}{2})},$$

where $\tilde{G}$, the binned block-maximal rank, recurs in the analysis below. The outer measure stays the true P ; only the surprisal *values* entering the rank are binned. This is what makes the error analysis of Section 6 rank-only.

Algorithm 2 – Compute s to certified binning error δ (general P)

Input: marginals $\{p_i\}$, target $\delta > 0$.

Output: estimate $\hat{s}$ and a certified bound $B \leq \delta$ on $|s_\eta - s|$.

- (1) Choose a bin width $\eta_0 \ll \min_i \text{gap}_i$, the smallest nonzero per-coordinate surprisal gap.
 - (2) Round each $s_i(a)$ strictly up to $\eta_0\mathbb{Z}$; form the mass and count histograms.
 - (3) Run Algorithm 1 on both chains to obtain $\tilde{\pi}, \tilde{F}$, and from them $f_\eta(1)$ and $f_\eta(\frac{1}{2})$ by (6).
 - (4) Assemble the certificate B on $|s_\eta - s|$ of Proposition 6.9 from $(\tilde{\pi}, \tilde{F}, \tilde{\mathcal{A}}_\rho)$, in one pass over arrays already in memory.
 - (5) If $B > \delta$, rerun steps 2–4 at $\eta_1 = \eta_0 \delta/B$, halving instead whenever that prediction under-delivers.
 - (6) Report $\hat{s}$, B , and the diagnostics Λ_η (10) and the empty-lattice fraction near the mean.
-

³Strictness — surprisals already on the grid are lifted by a full η rather than fixed — is what places the optimal rank in the window of Lemma 6.1 at no cost to the constants. Round-to-nearest remains admissible and typically does better — its offsets are approximately centered, so the signed combination $e_1 - s e_{1/2}$ of the moment errors is a difference of comparable terms, the effective window collapses from $m\eta$ to $\sqrt{m}\eta$, and one observes $|s_\eta - s| = O(\eta/\sqrt{m})$; this is a heuristic and no part of any guarantee. Its offsets are, however, two-sided, $|\tilde{S} - S| \leq m\eta/2$, and re-running the proof of Lemma 6.1 with the two-sided bound widens the window to $(m+1)\eta$, so the same $C(s)$ holds with ϵ_m increased by $1/m$.

Three of these steps carry a caveat worth stating plainly. The bin width in step 1 must resolve the per-coordinate structure: coarser than the gaps, binning collapses distinct levels and the certificate, while still valid, becomes vacuous. No a priori constant enters that choice — only the marginals, which are given. The prediction in step 5 is motivated by the scaling law’s linearity in η , but nothing rests on it: an over-optimistic η_1 simply returns a B still above δ and the schedule continues, so correctness never depends on the prediction, only termination speed does. And the diagnostics of step 6 must be recomputed at *this* η : the band $|t| \leq \pi/\eta$ widens as η shrinks, so a Λ_η computed at a coarser bin width does not transfer. The empty-lattice fraction is a rigorous falsifier: an empty grid point near the mean will contradict Lemma 6.3 whenever $\epsilon_\tau < 0.399$.

THEOREM 5.1 (PRECISION AND COST). (i) *Guarantee.* For every bin width η , the exact-arithmetic estimator satisfies $|s_\eta - s| \leq B$, where B is the computable certificate (13) of Proposition 6.9. The floating-point realization incurs in addition the error of Lemma 4.4; under directed-rounding interval arithmetic the resulting enclosure is rigorous end-to-end. (ii) *Scaling and cost.* Under the scaling-law hypotheses of Theorem 6.7, the estimator satisfies the multiplicative scaling law

$$|s_\eta - s| \leq C(s) \eta (1 + \epsilon_m), \quad C(s) = \frac{2 + s}{\tilde{H}_{2/3}}, \quad (7)$$

with ϵ_m computed as in Theorem 6.7, and $\epsilon_m = O(m^{-1/2})$ under the regularity conditions of Corollary 6.8; the returned bound tracks it, $B = O(C(s) \eta)$. Consequently, for uniformly bounded alphabets, the refinement schedule of Algorithm 2 reaches every target δ above the model’s atom-spacing floor (Remark 6.10) at final bin width $\eta_{\text{final}} = \Theta(\delta/C(s))$ and total cost $\tilde{O}(m/\eta_{\text{final}}) = \tilde{O}(m C(s)/\delta)$, the geometric schedule being dominated by its final run; the cost is independent of the surprisal range.

The leading constant is $\Theta(1)$ in m and free of the range R for any fixed per-coordinate law, but it is not bounded over families: it grows as $\tilde{H}_{2/3} \rightarrow 0$, and high-confidence advice — where a single symbol carries almost all the mass, placing large surprisals in the tail — can inflate it by more than an order of magnitude over a mildly biased coordinate. The cost $\tilde{O}(m C(s)/\delta)$ inherits that growth. The certificate route is indifferent to it: it reads its bound off the run, and a large $C(s)$ appears there as a larger returned B at the same η . The theorem is proved in the next section (both scaling law and certificate) together with Lemma 4.4 (floating-point error).

6 Accuracy of the binned estimator

We bound $|s_\eta - s|$ and prove Theorem 5.1. The strategy: an exact, CLT-free *sandwich* traps the true and the binned rank of every key in a *common* window of the binned (lattice) CDF $\tilde{F}$. The window is anchored at the true surprisal and extends (only) upward by the worst-case rounding shift $m\eta$; nevertheless the resulting error is sign-free since the relative order of the two ranks inside the window is uncontrolled. The sandwich thus reduces the binning error to a log-ratio of $\tilde{F}$ across that window (Section 6.1). We then use our tilting apparatus, equipped with the level-saddle and rate function, to identify the local growth rate of that lattice CDF (Section 6.2); a value-level estimate of the discrete reversed hazard

caps the rate over $\{t \geq H\}$ through the exact $|\tau_t| \leq 1$ of Corollary 6.5, with the Bahadur–Rao edge correction z_H^{-2} among the computed sub-dominant terms. Together these steps yield the two-sided worst-case bound (7). Beyond the hypotheses Theorem 6.7 carries explicitly, no further regularity is assumed. The scalars $\sigma_\tau, \beta_\tau, \Lambda_\eta$ and the localization gap enter as *computed inputs* rather than as hypotheses on their growth; asymptotic readings such as $\varepsilon_m = O(m^{-1/2})$ under Corollary 6.8 appear as interpretive corollaries, and R enters only through the level count $O(R/h)$. Where the computed scalars degrade — near-uniform or very high-confidence advice — the bounds report the degradation rather than excluding the case (Remark 6.10).

6.1 The lattice shift sandwich

Herein, the key step is to compare both ranks against the *binned* CDF $\tilde{F}$ (a genuine lattice law) so that the analysis never touches the true, generically non-lattice F_S except through one elementary, CLT-free inequality. A sandwich is genuinely needed because binning is *not* order-preserving: with $\eta = 1$, per-coordinate surprisals (1.0, 0.7) sum to 1.7 and round strictly up to 3, while (0.9, 0.9) sum to 1.8 and round to 2 — bins reorder keys, so no exactness argument is available in the general case and the window below does real work.

LEMMA 6.1 (LATTICE SHIFT SANDWICH). *For every key x , the true optimal rank interval $J(x) = [N \cdot F_S(S(x)^-), N \cdot F_S(S(x))]$ and the binned block $\tilde{J}(x) = [N \cdot \tilde{F}(\tilde{S}(x)^-), N \cdot \tilde{F}(\tilde{S}(x))]$ both lie in the common binned window $W(x) := [N \cdot \tilde{F}(S(x)), N \cdot \tilde{F}(S(x) + m\eta)]$. Hence, for every $\rho > 0$, we have $|\ln \tilde{\mathcal{A}}_\rho(\tilde{S}(x)) - \rho \ln G(x)| \leq \rho \Delta(x)$ with $\Delta(x) := \ln \frac{N \cdot \tilde{F}(S(x) + m\eta)}{\max(N \cdot \tilde{F}(S(x)), 1)}$, a log-ratio of the lattice $\tilde{F}$ floored at rank 1.*

PROOF. Strict rounding gives $0 < \tilde{S} - S \leq m\eta$ pointwise, which strengthens the CDF comparison: (a) if $\tilde{S}(y) \leq t$ then $S(y) < t$, so $\tilde{F}(t) \leq F_S(t^-)$; (b) if $S(y) \leq t$ then $\tilde{S}(y) \leq t + m\eta$, so $F_S(t) \leq \tilde{F}(t + m\eta)$. Thus

$$\tilde{F}(t) \leq F_S(t^-) \leq F_S(t) \leq \tilde{F}(t + m\eta) \quad \text{for all } t \quad (\text{CLT-free}). \quad (8)$$

Four endpoint comparisons, all at $t = S(x)$, now place both intervals in $W(x)$: the upper end of J by the right inequality of (8); the lower end of J by the *left* one, $N \cdot \tilde{F}(S(x)) \leq N \cdot F_S(S(x)^-)$ — the step that fails under non-strict rounding, and the only place strictness enters for the true rank; the upper end of $\tilde{J}$ since $\tilde{S}(x) \leq S(x) + m\eta$ and $\tilde{F}$ is non-decreasing; and the lower end of $\tilde{J}$ since $\tilde{S}(x) > S(x)$ forces $\{\tilde{S} \leq S(x)\} \subseteq \{\tilde{S} < \tilde{S}(x)\}$, i.e. $N \cdot \tilde{F}(S(x)) \leq N \cdot \tilde{F}(\tilde{S}(x)^-)$ — strictness again, and only here. For the log bound: $G(x) \in J(x)$ by Lemma 3.1(ii), and $\tilde{\mathcal{A}}_\rho(\tilde{S}(x))$ is an average of r^ρ over $r \in \tilde{J}(x)$, so both $G(x)^\rho$ and $\tilde{\mathcal{A}}_\rho(\tilde{S}(x))$ lie between the ρ -th powers of the endpoints of $W(x)$; since both are ranks — positive integers — the lower endpoint may be floored at 1, which keeps Δ finite for *every* key: under strict rounding $\tilde{F}(\min S) = 0$ always (every $\tilde{S}$ exceeds $\min S$), so without the floor Δ would be undefined at the minimum-surprisal key. Taking logs gives the bound. $\square$

The floor is never active where the analysis uses the sandwich: on $\{S \geq H\}$ one has $N \cdot \tilde{F}(H) \geq 1$ as soon as $H - \min S =$

$\sum_i (H_1(p_i) - H_\infty(p_i)) \geq m\eta$ — the left side is $\Theta(m)$ whenever the advice is non-flat on a constant fraction of coordinates, the same non-degeneracy Theorem 6.7 invokes, while $m\eta = O(\sqrt{m})$ in the scaling regime; the excluded flat case has P uniform on its support, outside standing assumption (iv). The identical positivity requirement appears in Lemma 6.6's ratio $\tilde{v}(t)/\tilde{F}(t)$ and is discharged by the same inequality.

The size of $\Delta(x)$ is set by how fast $\tilde{F}$ grows across a window of width $m\eta$ at the level $t = S(x)$, i.e. the *rank-growth rate*. Routing through $\tilde{F}$ makes the entire accuracy argument lattice-only, and the sole Fourier-side input is the computed lattice-span exponent Λ_η of Lemma 6.3 — a hypothesis of the scaling law only, not of the computable certificate (Proposition 6.9).

Remark 6.2 (sublattices, resonances, and the computed test). If all binned surprisals lie on a common sublattice of span $k\eta$ (a genuine *coarse sublattice*), nothing degrades: the leading $C(s)\eta$ is set by the rounding resolution regardless of where the binned values land, and the local CLT applies at span $k\eta$ instead — one takes the gcd of the binned surprisals to identify the true span and computes Λ_η on the matching band. What the gcd test cannot see is *nearly-commensurate* advice, and this is neither undetectable nor an edge case: it is precisely the event $\Lambda_\eta \ll m$, and it is detected by computing Λ_η . A concrete instance with bounded leakage and gcd = 1: at $\eta = 0.01$, take 90% of coordinates Bernoulli(0.8) and 10% Bernoulli(0.5025) — binned gaps of 138 and 1 lattice steps, so the summed lattice genuinely has span η ; yet $\Lambda_\eta = 0.0036$ at $m = 64$, and 94% of the bulk lattice points carry *exactly* zero mass, so the span- η local CLT's conclusion is false there. Neither gcd = 1 nor bounded per-coordinate leakage implies the hypothesis; only the computed Λ_η decides it. At the opposite extreme, advice on a *proper* coarse sublattice, $h = k\eta$ with $k \geq 2$, forces $\Lambda_\eta = 0$: the resonance $t = 2\pi/h$ then lies inside the band $|t| \leq \pi/\eta$, where every $\sin^2(t\Delta_{iab}/2)$ vanishes — so the scaling law's hypothesis fails where Theorem 4.1 applies. This is no loss: those inputs belong to the exact engine (Section 4.1), not the binned route. In every case the certificate of Proposition 6.9 is unaffected; only the scaling law reads Λ_η .

6.2 The level-saddle apparatus and the rate

Lemma 6.1 reduced the binning error to a single quantity: the growth $\ln[\tilde{F}(t + m\eta)/\tilde{F}(t)]$ of the lattice CDF across the rounding window. Both entries are lower-tail probabilities of size $e^{-\Theta(m)}$, so we compute the ratio by exponential tilting, i.e. reweighting U after which the rare level t becomes typical, so that the ratio is a bulk quantity governed by a central limit theorem rather than a tail estimate. Two objects attached to the uniform CGF Λ_U run this programme. For $t < \mu_U$ the *saddle* $\tau_t < 0$ is the unique root of $\Lambda'_U(\tau_t) = t$: the tilted law $U_{\tau_t}(x) \propto e^{\tau_t S(x)}$ has mean surprisal exactly t . As t increases, τ_t increases with it, which means that $t \mapsto |\tau_t|$ is strictly decreasing. The rate function $I_U(t) = \sup_\tau (\tau t - \Lambda_U(\tau))$, the convex conjugate of Λ_U , prices the tail and satisfies $I'_U(t) = \tau_t$.⁴

⁴For $\tau < 0$, Markov's inequality applied to $e^{\tau S}$ gives the Chernoff bound $F_S(t) \leq e^{-(\tau t - \Lambda_U(\tau))}$; the supremum selects the best exponent and is attained at the saddle, $I_U(t) = \tau_t t - \Lambda_U(\tau_t)$. The exponent is tight: the tilting identity (9) below isolates the missing prefactor exactly (the term $M(0)$), which the local CLT evaluates to $\Theta(m^{-1/2})$ throughout the saddle region, giving $F_S(t) = e^{-I_U(t)} \Theta(m^{-1/2})$. Finally,

A one-line change of measure ($dU/dU_{\tau_t} = e^{-\tau_t S + \Lambda_U(\tau_t)}$) makes the tail–bulk exchange exact: with $\tau = \tau_t$ and $\tilde{S} = S - t$, centered under U_{τ_t} ,

$$F_S(t + a) = e^{-I_U(t)} M(a), \quad M(a) := \mathbb{E}_{U_{\tau_t}}[e^{|\tau_t|\tilde{S}}; \tilde{S} \leq a], \quad (9)$$

(9) factors the tail probability into the large-deviation cost $e^{-I_U(t)}$, which carries the entire exponential t -dependence, and a window term $M(a)$ whose weight $e^{|\tau_t|\tilde{S}}$ is the likelihood ratio left over after extracting the cost at the anchor t . Since the tilted point masses are nearly flat near the mean while the weight decays by $e^{-|\tau_t|\eta}$ per lattice step down, $M(a)$ is a geometric-type sum dominated by the $O(1/|\tau_t|)$ of surprisal just below $t + a$, which is the origin of the growth rate $|\tau_t|$ below. In the ratio $F_S(t + m\eta)/F_S(t) = M(m\eta)/M(0)$ the cost cancels exactly, leaving a quantity governed by the tilted law near its mean; the same identity holds verbatim for the binned chain. The one analytic input is a lattice local CLT for the tilted law at the density level: M probes individual point masses, so CDF-level control would not suffice (also see Remark C.1).

LEMMA 6.3 (TILTED/LATTICE LOCAL CLT, EXPLICIT). *Under U_τ , $\tau \in [-1, 0]$, let $\tilde{S} = \sum_i (Y_i - \mathbb{E}Y_i)$ with $Y_i = \tilde{s}_i(X_i)$ the binned per-coordinate surprisals: a centered lattice variable of span η (on a grid offset by the centring), with variance σ_τ^2 and summed third moment $\beta_\tau = \sum_i \mathbb{E}_{U_\tau} |Y_i - \mathbb{E}Y_i|^3$. Write $g_k = p_k/\eta$ with $p_k = \mathbb{P}_{U_\tau}[\tilde{S} = s_k]$, let ψ_i be the characteristic function of the centered Y_i , and set $\delta_1 := \min\{3\sigma_\tau^2/(2\beta_\tau), 1/\max_i \sigma_i\}$. Define the lattice-span exponent*

$$\Lambda_\eta := \min_{\delta_1 \leq |t| \leq \pi/\eta} \sum_i \sum_{a < b} 4 p_{i,a}^{(\tau)} p_{i,b}^{(\tau)} \sin^2(t \Delta_{iab}/2), \quad (10)$$

(if $\delta_1 \geq \pi/\eta$ the band is empty, $\Lambda_\eta := +\infty$, and the high-frequency contribution vanishes), with Δ_{iab} the binned surprisal gaps of coordinate i and $p_{i,a}^{(\tau)}$ the tilted weights. Then $|\prod_i \psi_i(t)| \leq e^{-\Lambda_\eta/2}$ on the band, and $\sup_k |g_k - \frac{1}{\sigma_\tau} \varphi(s_k/\sigma_\tau)| \leq \frac{\varepsilon_\tau}{\sigma_\tau}$ with $\varepsilon_\tau = \frac{4}{3\pi} \frac{\beta_\tau}{\sigma_\tau^3} + q_\tau + \frac{\sigma_\tau}{\eta} e^{-\Lambda_\eta/2} + \sqrt{\frac{2}{\pi}} \bar{\Phi}(\sigma_\tau \delta_1)$, where q_τ is the quartic correction (a computed integral over the marginals, see Appendix B). Every term of ε_τ is a computed number; under the regularity conditions of Corollary 6.8 the asymptotic reading is $\varepsilon_\tau = \frac{4}{3\pi} \beta_\tau/\sigma_\tau^3 + \Theta(1/m) + O(e^{-\Theta(m)}) = \Theta(m^{-1/2})$ (Corollary 6.8). φ denotes the standard normal density.

The full proof is in Appendix B. The intuition: after tilting, the histogram of $\tilde{S}$ tracks the Gaussian curve pointwise. Each lattice point carries mass $p_k \approx \eta \varphi(s_k/\sigma_\tau)/\sigma_\tau$ with an error uniform in k ; hence the relative error is $O(m^{-1/2})$ at points within $O(\sigma_\tau)$ of the mean, precisely the points dominating the sums $M(a)$. Three usage notes. (a) Only the scaling law (Theorem 6.7, Lemmas 6.4 and 6.6) consumes this lemma; the computable certificate (Proposition 6.9) never touches it. (b) Computing ε_τ shows where the lemma has content: on healthy heterogeneous Bernoulli advice $\varepsilon_\tau > 1$ at $m = 32$ for every η (the high-frequency term dominates), it is η -dependent at $m = 64$ (0.18 at $\eta = 0.1$, 4.9 at $\eta = 10^{-3}$), and only from $m \gtrsim 128$ is it comfortably below one (0.070 at $m = 256$, $\eta = 10^{-3}$). The asymptotic reading is correct but takes hold only at large m , which is why the certificate route does not rely on it. (c) The scalars are quantified over the tilt range $\tau \in [-1, 0]$ that

differentiating $I_U(t) = \tau_t t - \Lambda_U(\tau_t)$ and using $\Lambda'_U(\tau_t) = t$ cancels the $d\tau_t/dt$ terms (the envelope theorem), leaving $I'_U(t) = \tau_t$: the local slope of the tail exponent is the saddle itself, already foreshadowing the rank-growth rate of Lemma 6.4.

the engine uses; across the tested ensembles the supremum of the Berry–Esseen term sits at the edge tilt $\tau = -1$ – the tilt in force at the dominant level $t = H$ (Corollary 6.5) – so uniformity over the range costs a single evaluation.

At exponential order the rate is forced by duality: $\ln F_S(t) \approx -I_U(t)$ and $I'_U(t) = \tau_t$, so the log-CDF climbs at slope $|\tau_t|$. Ranks grow by a factor of $e^{|\tau_t|\Delta t}$ per surprisal increment Δt . The next lemma makes the heuristic exact up to an explicit relative error at finite m , uniformly across the rounding window.

LEMMA 6.4 (RANK-GROWTH RATE, MULTIPLICATIVE FORM). *Fix t with $|\tau_t| = \Theta(1)$ (the saddle region, covering the left edge $t = H$ where $|\tau_H| = 1$). Write $u_t := m\eta/\sigma_t$ and $z_t := |\tau_t|\sigma_t$. Provided $u_t \leq u_0$ for a fixed small constant u_0 and under the computed small-error condition $c_1 \varepsilon_\tau \leq c_0 < 1$, for the absolute constant c_1 of the proof (hypotheses of this sharp form and of the later Theorem 6.7 only)*

$$\ln \frac{\tilde{F}(t + m\eta)}{\tilde{F}(t)} = |\tau_t| m\eta (1 + O(u_t/z_t) + O(z_t^{-2}) + O(\varepsilon_\tau)),$$

with ε_τ from Lemma 6.3 and the constants depending only on u_0 ; the leading coefficient is the local rank-growth rate $\theta(t) = |\tau_t|$. Under Corollary 6.8, $u_t/z_t = \Theta(\eta)$ and the relative correction is $O(m^{-1/2})$.

The proof (Appendix C) substitutes Lemma 6.3 into (9): $M(m\eta)$ and $M(0)$ each split into a Gaussian surrogate and a local-CLT residual sharing the same envelope. The Gaussian ratio is an integral of the Gaussian hazard rate over an interval of width $m\eta/\sigma_\tau$ and evaluates to $|\tau_t|m\eta$ up to relative $O(u_t/z_t) + O(z_t^{-2})$; the shared envelope makes the residual a further relative $O(\varepsilon_\tau)$. Bounding numerator and denominator separately would instead leave an additive error independent of the window width, creating a floor on the certifiable precision that survives $\eta \rightarrow 0$. Differencing the logarithms first turns it into a relative correction, so no such floor remains.

COROLLARY 6.5 (THE EXACT RANK-GROWTH RATE). *For every non-degenerate product posterior (standing assumption (iv), $\text{Var}_U(S) > 0$), $\theta^* := |\tau_H| = 1$, where $\Lambda'_U(\tau_H) = H$. Hence $\theta(t) = |\tau_t| \leq \theta^* = 1$ for $H \leq t \leq \mu_U$; beyond μ_U the rate is covered by the outside-saddle estimate of Lemma 6.6.*

PROOF. $U_\tau^{(i)}(a) \propto e^{\tau s_i(a)} = p_{i,a}^{-\tau}$ gives $U_{-1} = P$ and $\Lambda'_U(-1) = \mathbb{E}_P[S] = H$ unconditionally; non-degeneracy makes Λ_U strictly convex, so the root of $\Lambda'_U(\tau) = H$ is unique, whence $\tau_H = -1$ and $\theta^* = 1$. The saddle levels dominating $f(\rho)$ ($\frac{1}{2} \leq \rho \leq 1$) satisfy $t_\rho^* \geq H$, since π is centered at H and F_S^ρ is nondecreasing; since Λ'_U is increasing with $\Lambda'_U(-1) = H$ and $\Lambda'_U(0) = \mu_U$, $|\tau_t|$ decreases from 1 to 0 across $[H, \mu_U]$, which gives the bound there. $\square$

This $\theta^* = 1$ is the asymptotic-equipartition statement [13] that $G^* \approx 1/P = e^S$ for typical keys ($d \ln G^*/dS = 1$ at $S = H$). The tempting Gaussian estimate Γ/σ_U^2 is only a first-order approximation of it: by the mean value theorem we have $\Gamma/\sigma_U^2 = 1 + O(\Gamma \Lambda_U''/\sigma_U^4)$, a $\Theta(1)$ correction of either sign, therefore we use the exact value 1 instead of the Gaussian proxy.

6.3 The edge term and the worst-case bound

Lemma 6.4 gave the growth rate at one fixed level; the worst-case bound needs a bound over all levels a contributing key can occupy.

However, only $\{t \geq H\}$ really matters (Theorem 6.7 shows the sub-entropy region carrying exponentially little weight), and there the exact convexity inequality $|\tau_t| \leq |\tau_H| = 1$ of Corollary 6.5 caps the rate outright: the worst key for binning is the *typical* one. The next lemma is a *value-level* estimate of the single-step discrete reversed hazard $\tilde{v}(t)/\tilde{F}(t)$ — no derivative of any expansion is taken, and no monotonicity of a profile is claimed; the supremum over $\{t \geq H\}$ is located by Corollary 6.5 alone. Intuitively, two effects drive the rate: a large-deviation exponent contributes the slope $|\tau_t|$, and the saddlepoint prefactor contributes the relative correction z_t^{-2} with $z_t = |\tau_t|/\sigma_t$; at the edge $t = H$ the latter is the Bahadur–Rao correction [5] $z_H^{-2} = 1/(m\text{Var}_1)$, with Var_1 the mean per-coordinate surprisal variance under P — sub-dominant beside the local-CLT term ε_t under Corollary 6.8’s regularity, which is why it appears here in prose rather than in the boxed bound.

LEMMA 6.6 (DISCRETE REVERSED HAZARD, VALUE-LEVEL). *For $t \geq H$ in the saddle range $|\tau_t| = \Theta(1)$, and under the computed small-error condition $\sqrt{2\pi} \varepsilon_t \leq c_0 < 1$, the binned law satisfies $\tilde{v}(t)/\tilde{F}(t) \leq |\tau_t| \eta (1 + \varepsilon_t(t))$, with $\varepsilon_t(t) \leq 2\sqrt{2\pi} \varepsilon_t (1 + O(\varepsilon_t)) + O(z_t^{-2}) + O(|\tau_t|/\eta) + O(|\tilde{\tau}_t - \tau_t|/|\tau_t|)$ where ε_t is from Lemma 6.3. Every scalar in ε_t is computed and the constants are absolute; outside the saddle range ($z_t = O(1)$) the ratio is $O(\eta/\sigma_t)$, strictly smaller. Consequently, with $|\tau_t| \leq 1$ on $[H, \mu_U]$ (Corollary 6.5) and the smaller ratio beyond, we have $\sup_{t \geq H} \tilde{v}(t)/\tilde{F}(t) \leq \eta (1 + \varepsilon_t)$.*

The proof (Appendix C) reads the single-step ratio directly off the tilting identity and Lemma 6.3’s point-mass control — a value statement, which is all the local CLT can support: value control of an approximation gives no derivative control ($m^{-1/2} \sin(mt)$ is $O(m^{-1/2})$ with derivative $\Theta(m^{1/2})$). The correction ε_t carries z_t^{-2} explicitly rather than $O(m^{-1/2})$, so no implicit $z_t \gg 1$ assumption remains and the near-uniform regime is covered by the same statement.

THEOREM 6.7 (BINNING ERROR — THE SCALING LAW). *Assume the scaling-law hypotheses: the regime $\sqrt{m} \eta \leq \varepsilon_0$ for a fixed small constant ε_0 ; the lattice-span condition $\inf_{\tau \in [-1,0]} \Lambda_\eta(\tau) \geq \lambda m$ (Lemma 6.3); the entropy gap $\tilde{H}_{1/(1+\rho)} - \tilde{H}_1 \geq g > 0$ for $\rho \in [\frac{1}{2}, 1]$ (advice non-uniform on a constant fraction of coordinates); $\log(1 + \ln N) = o(m)$, so that Arkan’s additive slack is lower-order; and the computed small-error condition $\sup_{\tau \in [-1,0]} \sqrt{2\pi} \varepsilon_\tau \leq c_0 < 1$ of Lemma 6.3. Then, with $\theta^* = 1$ (Corollary 6.5) and $f(\frac{1}{2}) = \frac{1}{2} m \tilde{H}_{2/3} (1 + o(1))$ from (1), we have $|s_\eta - s| \leq \frac{2+s}{\tilde{H}_{2/3}} \eta (1 + \varepsilon_m)$, which is the scaling law behind (7); ε_m reduces to computed scalars with absolute constants, through Lemma 6.3, by the propagation $\varepsilon_m \leq 2\sqrt{2\pi} \varepsilon_t (1 + O(\varepsilon_t)) + \bar{z}^{-2} + O(\eta) + O(\log(1 + \ln N)/f_\eta(\frac{1}{2}))$ (Appendix C). Under the regularity conditions of Corollary 6.8, $\varepsilon_m = O(m^{-1/2})$. None of these hypotheses is consumed by the computable certificate (Proposition 6.9), which bounds $|s_\eta - s|$ from the run’s own arrays.*

The constant is transparent: a key’s log-rank moves by at most $\theta^* m \eta = m \eta$ (rate $\times$ window), so each moment exponent $f(\rho)$ moves by at most $\rho m \eta$; the ratio $s = f(1)/f(\frac{1}{2})$ divides by $f(\frac{1}{2}) = \frac{1}{2} m \tilde{H}_{2/3} (1 + o(1))$, the m ’s cancel, and what survives is the per-coordinate resolution η measured in units of the per-coordinate entropy $\tilde{H}_{2/3}$. The factor $2 + s$ merely collects the numerator and

denominator contributions ($e_1 - s e_{1/2}$ in the proof). (Proof in Appendix C.)

COROLLARY 6.8 (INTERPRETIVE SCALING). *If the alphabets are uniformly bounded, the per-coordinate leakage is bounded, $\max_a s_i(a) - \min_a s_i(a) \leq c$ for all i , and the range is extensive, $R = \Theta(m)$, then uniformly over the tilt range $\tau \in [-1, 0]$: $\sigma_\tau^2 = \Theta(m)$, $\beta_\tau \leq c \sigma_\tau^2$, hence $\beta_\tau/\sigma_\tau^3 = O(m^{-1/2})$ (Lyapunov); if moreover $\Lambda_\eta \geq \lambda m$ and $\log(1/\eta) = o(m)$, then $\varepsilon_\tau = \Theta(m^{-1/2})$ and $\varepsilon_m = O(m^{-1/2})$. (Proof in Appendix C.) This is where the familiar regularity conditions live in this paper: as an interpretation of computed scalars, not as a gate on any guarantee.*

A tighter, ρ -localized constant follows from replacing θ^* by the rate $|\tau_{t_\rho}^*| < 1$ at the saddle level $t_\rho^* > H$ dominating $\mathbb{E}_\rho[G^\rho]$. Note also what the bound does *not* contain: binning perturbs only the ranks, never the weights — the expectation is still taken under the true P , and N is unchanged — so Lemma 6.1 already captures the entire perturbation, and $C(s)$ needs no additional term for the measure.

6.4 The computable certificate

The scaling law explains the error; it does not let a reader holding the marginals and m evaluate it. But every asymptotic ingredient above is a functional of $\tilde{F}$ — and $\tilde{F}$ is an output. The growth rate that Lemma 6.6 estimates can be *measured* on the computed $\tilde{F}$; the denominator that (1) approximates is, up to $e_{1/2}$, the computed $f_\eta(\frac{1}{2})$; and the localization threshold can be dispensed with level by level. The result is an a-posteriori certificate in the style of validated numerics: an interval, not a formula.

PROPOSITION 6.9 (COMPUTABLE CERTIFICATE). *Run the engine at bin width η with strict upward rounding and let $(\tilde{\pi}, \tilde{F}, \tilde{\mathcal{A}}_\rho)$ be its arrays. Under the truncated merge of Lemma 4.3(ii) the arrays carry a one-sided mass deficit, and $\tilde{F}$ is replaced throughout by the computed enclosure $[\tilde{F}(t), \tilde{F}(t) + \varepsilon_{\text{tr}} e^{|\phi|t}]$ of Appendix E, with ε_{tr} the truncation tolerance defined there, its lower end in lo_ρ and its upper end in hi_ρ , which leaves (12) valid. For each level t of $\tilde{S}$ define*

$$\begin{aligned} D(t) &:= \max_{u \in \{t-m\eta, \dots, t-\eta\}} \ln \frac{\tilde{F}(u + m\eta)}{\tilde{F}(u)} \quad (\infty \text{ if some } \tilde{F}(u) = 0), \\ \text{lo}_\rho(t) &:= \max \left\{ (N \cdot \tilde{F}(t - m\eta))^\rho, 1, \tilde{\mathcal{A}}_\rho(t) e^{-\rho D(t)} \right\}, \\ \text{hi}_\rho(t) &:= \min \left\{ (N \cdot \tilde{F}(t + m\eta))^\rho, e^{\rho t}, \tilde{\mathcal{A}}_\rho(t) e^{\rho D(t)} \right\}, \end{aligned} \quad (11)$$

dropping the D -candidates where $D(t) = \infty$. Then

$$\sum_t \tilde{\pi}(t) \text{lo}_\rho(t) \leq e^{f(\rho)} \leq \sum_t \tilde{\pi}(t) \text{hi}_\rho(t), \quad (12)$$

and, with E_ρ the induced computed bound on $|e_\rho| = |f_\eta(\rho) - f(\rho)|$,

$$|s_\eta - s| \leq B := \frac{E_1 + s_\eta E_{1/2}}{f_\eta(\frac{1}{2}) - E_{1/2}}, \quad \text{valid whenever } f_\eta(\frac{1}{2}) > E_{1/2}. \quad (13)$$

Every quantity on the right is computed in one pass over arrays already in memory. The proposition uses only the engine assumptions and strict rounding: no local CLT, no lattice-span condition, no regime

constant ε_0 , no saddlepoint expansion, and no Arikan scale (proofs in Appendix D).

The three candidate bounds per level are complementary: the rank window survives at the extremes of the lattice, the Chernoff cap $e^{\rho t}$ handles the top of the distribution, and the D -window is the tight one through the bulk. Carrying e_ρ as a signed interval rather than the symmetric E_ρ tightens the result further, the two sides of (12) being asymmetric. Measured against exact ground truth on exhaustively enumerable models ($m \leq 10$): the bracket holds throughout, the certificate sits within a factor 1.14–1.45 of the leading term $C(s)\eta$ for $\eta \leq 0.1$, closing monotonically with m and η , and on heterogeneous advice it scales cleanly linearly in η across seven halvings with no floor in sight. Since (13) is a theorem about exact arithmetic while the arrays are floating point, it certifies the binning error of the exact estimator. The additional floating-point error is controlled by Lemma 4.4, but the reference implementation does not provide a rigorous end-to-end numerical enclosure. Running the same computation in directed-rounding interval arithmetic (Section 4.2) would provide such an enclosure and remove the remaining unnamed polynomial. The exact-arithmetic certificate is valid under the engine assumptions alone, and informative exactly where the level structure is rich enough to measure (Remark 6.10).

Having Proposition 6.9, Theorem 6.7, Lemma 4.4 and Lemma 4.3, we are now set to prove Theorem 5.1:

PROOF OF THEOREM 5.1. (i) By the triangle inequality, $|\widehat{s} - s| \leq |s_\eta - s|_{\text{bin}} + |\widehat{s} - s_\eta|_{\text{fp}}$; Proposition 6.9 bounds the first term by the computed certificate B , while Lemma 4.4 separately bounds the floating-point term. Under the interval-arithmetic route both contributions can be enclosed rigorously end-to-end. No hypothesis beyond the engine assumptions enters. (ii) The scaling law (7) is precisely Theorem 6.7. For the tracking claim: under Theorem 6.7’s hypotheses, every contributing level has $D(t) \leq m\eta(1 + \varepsilon_m)$ by Lemma 6.6 and the telescoping argument, while the finitely many levels with $D(t) = \infty$ are covered by the rank-window and Chernoff candidates of (11) and contribute the exponentially small localization remainder; hence $E_\rho \leq \rho m\eta(1 + \varepsilon_m) + O(e^{-\Theta(m)})$ and (13) gives $B \leq C(s)\eta(1 + \varepsilon_m)(1 + o(1))$. A run at η therefore certifies binning error δ once $\eta = \Theta(\delta/C(s))$; the geometric fallback of Algorithm 2 halves η at most $O(\log(1/\delta))$ times, for uniformly bounded alphabets, each run costs $\tilde{O}(m/\eta)$ (Lemma 4.3), and the final run dominates. $\square$

Remark 6.10 (status of the guarantees, and the floors). Precisely which statement is non-asymptotic. (i) Theorem 4.1 is an exact identity with no discretization error: s is a ratio of finite level sums, and the only numerical error is the floating-point evaluation covered by Lemma 4.4. “Exact” thus refers to the underlying finite-sum identity, not exact numerical representability. (ii) Proposition 6.9 is non-asymptotic and computable: in exact arithmetic, B is a rigorous bound on $|s_\eta - s|$ under the engine assumptions alone. (iii) Theorem 6.7 is an asymptotic scaling law with computed constants — the interpretive statement explaining why refining η works and where $C(s)$ comes from — and it is the only place where its hypotheses enter. As for floors on the reachable precision: the certificate’s floor is the *atom spacing* of the binned surprisal law. For heterogeneous advice (rank $\Theta(m)$, Section 4.1) the spacing is $e^{-\Theta(m)}$ and this floor

is irrelevant — in practice the floating-point floor $\delta \sim 10^{-10}$ of Appendix E binds first — while for k -symbol i.i.d. advice the spacing is $\Theta(m^{1-k})$ and the certificate stalls near it; but that is exactly the finite-rank regime where the exact engine applies and no certificate is needed. One caution in the opposite direction: at finite m the substitution $f(\frac{1}{2}) \approx \frac{1}{2}m\tilde{H}_{2/3}$ of (1) overstates the true $f(\frac{1}{2})$, by $\approx 20\%$ at $m = 8$, and it sits in the denominator; the $O(\log(1 + \ln N)/f_\eta(\frac{1}{2}))$ term of ε_m is what keeps $C(s)\eta(1 + \varepsilon_m)$ conservative, and the computed certificate, which divides by the computed $f_\eta(\frac{1}{2})$, avoids it entirely.

Remark 6.11 (which engine to use, which guarantee to state—a user’s map). (1) *Level structure.* If the surprisals are commensurate or, more generally, of bounded rank d — a model hypothesis, like commensurability itself (Section 4.1) — the level count is $O(m^d)$: run the exact engine (Algorithm 1). No certificate is needed there, and the scaling law’s lattice hypothesis would in fact fail ($\Lambda_\eta = 0$, Remark 6.2). Otherwise, bin and run Algorithm 2. (2) *Certificate.* Each binned run returns s_η with the bound B of Proposition 6.9 on $|s_\eta - s|$, valid under its engine assumptions alone; refine η until $B \leq \delta$. Two floors bind: the atom spacing of the binned law ($e^{-\Theta(m)}$ for heterogeneous advice) and, in the floating-point realization, $\sim 10^{-10}$ (Appendix E). The scaling law (7) explains the observed $B = O(C(s)\eta)$ under its own hypotheses but is never load-bearing (Remark 6.10).

7 Cryptographic applications

We report the information-theoretic exponent $s = f(1)/f(\frac{1}{2})$ of Section 2 and nothing beyond it: Montanaro’s constant and additive overhead shift $\ln W_Q$ by $O(1)$, outside the exponent by construction, as in [7]. One rule keeps the classical baseline honest — *attack seeds or black boxes, not structured keys*. Where the secret is algebraic (NTRU ternary [22], ML-KEM binomial [9], LWE [29], LPN [2, 8], as catalogued in [7]), full-key ordered enumeration is not the optimal classical attack: decoding can replace it outright [8], and lattice reduction absorbs all but a guessed sub-block of the secret [26], though the exact efficiency of the latter is contested [14]. An exponent measured against full-key ordered guessing would therefore flatter the quantum side. We instantiate only where likelihood-ordered enumeration is provably the optimal classical attack.

(A) *Cold-boot advice.* DRAM cells do not lose their contents the instant power is cut; they decay towards a chip-dependent ground state, so an attacker who reboots and dumps memory recovers a noisy copy of whatever was held there [19]. Bits flip independently and asymmetrically, $\alpha = \mathbb{P}(0 \rightarrow 1) \ll \beta = \mathbb{P}(1 \rightarrow 0)$: an exact binary product posterior. Conditioning on the dumped bit under a uniform prior gives two marginal types, $\mathbb{P}(s^*=1 \mid \tilde{s}=1) = \frac{1-\beta}{1-\beta+\alpha}$ and $\mathbb{P}(s^*=1 \mid \tilde{s}=0) = \frac{\beta}{\beta+1-\alpha}$. We report the *expected* composition, giving the first type to $\lfloor W f_1 \rfloor$ of the W coordinates, $f_1 = \mathbb{P}(\tilde{s}=1) = \frac{1-\beta+\alpha}{2}$; the advice is then a function of (α, β, W) alone.⁵ Two unrelated secrets sit here: the short seeds from which many post-quantum signatures expand their entire private key [17], and block-cipher keys [6]. At equal length they carry identical advice. Neither admits

⁵The composition is the only free convention here. Over the binomial sampling distribution of the observed bit counts, s lies in $[3.77, 4.11]$ for AES-128 at $\beta = 0.01$ and in $[2.70, 2.85]$ at $\beta = 0.05$ (95%), with the reported value at the mean.

an algebraic shortcut, so ordered enumeration is the classical attack; and for the ciphers the corresponding Grover oracles have actually been built and costed, tolerating up to 40% noise [6, 17]. The cipher rows below follow that work’s channel and targets; its GIFT-128, LowMC-128 and Picnic-L1/L3/L5 instances are not listed separately because of the shared advice properties. The marginals have rank 2, so the finite-rank engine of Section 4.1 applies and the entries are exact.

(B) *Side-channel analysis of a hash PRF.* ML-KEM derives its encryption coin, and ML-DSA its masking token ρ' , by expanding a short secret through SHAKE-256. Recovering that seed breaks the scheme without touching its lattice problem, and because the seed is a structureless random string, enumeration is again the only classical route. The attack of [23] is a *soft-analytical* side-channel analysis, and its two-stage structure is what matters for us. First, a per-bit classifier reads the power trace and guesses each bit of the Keccak- f state independently; taken alone this is weak, and its quality is what that work tabulates as an *accuracy*. Second, what constitutes the actual attack is that those crude guesses are placed on a factor graph whose nodes are the state bits and whose edges are the permutation’s own XOR, AND and NOT relations, and *belief propagation* passes them around that graph until they agree. Thousands of algebraic constraints thereby correct one another, and the marginals that come *out* are sharper than the classifier that went *in* by many orders of magnitude. So the tabulated accuracies are the input to the attack, not its output, and cannot serve as our advice; what the attack reports as its result is the residual key *rank*, the number of candidates a classical attacker must still enumerate, which we use to calibrate a synthetic posterior model. We therefore fit a per-bit $\text{Ber}(q)$ advice reproducing the reported rank. This is a synthetic i.i.d. surrogate calibrated to that residual rank, not a measured posterior: the true post-propagation marginals are heterogeneous and, on a graph with cycles, not exactly independent.

(C) *Template attacks.* A template attack [11] is the strongest form of profiled side-channel analysis. The adversary first characterizes a device identical to the target, building for each of the 256 values a key byte may take a statistical model of the leakage it produces, and then matches a single trace from the target against those models. Since AES handles its key one byte at a time in the first round, the leakage is modeled as sixteen independent per-byte posteriors, the standard assumption here and a product advice of exactly the form our framework assumes, after which enumerating the candidates in likelihood order is the established classical procedure [32], whose guessing entropy is the standard yardstick for evaluating such attacks [31]. It is also the natural stress test: a real-valued posterior, simulated from a Gaussian leakage model at SNR anchors taken from the literature, whose surprisals are not commensurate, so it must be binned.

What the numbers show. The exponent is substantially super-quadratic wherever the advice is skewed — above 2.7 across the practical cold-boot range, and up to 3.97 in the synthetic model calibrated to the reported ML-DSA residual rank, an exponent of that model rather than of the unknown belief-propagation posterior — so our results suggest that there are cryptographic settings where choosing Montanaro’s algorithm over plain Grover is meaningful.

Table 2: Computed exponents against the bound of [7] (their Theorem 4.3). Block A is exact (finite-rank engine, Section 4.1), Block B is exact within the synthetic i.i.d. Bernoulli models fitted to the reported residual ranks, C is a binned setting. The † symbol marks values below 2.

setting	param	s	BGMN Thm 4.3
<i>A. cold boot, asymmetric channel $\alpha = 0.001$</i>			
AES-128, 128-bit seed	$\beta = 0.01$	3.918	2.310
	$\beta = 0.05$	2.763	2.187
	$\beta = 0.25$	2.199	1.975 †
AES-256, 256-bit seed	$\beta = 0.01$	3.639	2.757
	$\beta = 0.05$	2.713	2.387
	$\beta = 0.20$	2.245	2.101
PRESENT-80	$\beta = 0.05$	2.819	1.979 †
	$\beta = 0.20$	2.263	1.895 †
<i>B. synthetic Bernoulli models fitted to Keccak SASCA residual ranks</i>			
ML-KEM coin (256 b)	2^{69}	3.114	2.684
	2^{128}	2.513	2.336
ML-DSA ρ' (512 b)	2^{69}	3.974	3.301
	2^{128}	3.179	2.901
<i>C. AES template attack, 16 bytes over 256 values</i>			
unprotected 8-bit	SNR = 1	2.018	1.900 †
	SNR = 5	2.032	1.904 †

The prior bound [7] is loosest exactly there, and in five rows drops below 2 which certifies nothing at all: its additive penalty $\log(1 + \log |K|)$ is *fixed* while $H_{1/2}$ shrinks under strong leakage, which is also why the shortest key fares worst. Two smaller effects are worth recording. The corrected level averages $\mathcal{A}_\rho$ of Lemma 3.1 raised every entry of Table 2 above its block-maximal surrogate, so ignoring ties would have understated the advantage here; the sign is not fixed a priori (Remark E.1). And the asymmetric channel of block A yields a larger exponent than a symmetric one at equal β (2.763 against 2.532 at $n = 128$, $\beta = 0.05$), because it is the more skewed of the two.

Block C is the instructive negative. The exponent will not move much, staying at 2.011 even at a residual 0.54 bits per byte, sharper than a measured unprotected device [12]; Proposition 6.9 and the universal $s \geq 2$ return $s \in [2, 2.044]$ at SNR = 1. Uniform advice at this $N = 2^{128}$ already gives $s = 2.0027$, so the advice-driven excess is nearer 0.015 than 0.018. A Gaussian-shaped posterior over a large alphabet is *smooth*, so $\bar{H}_{1/2}/\bar{H}_{2/3} \approx 1.015$ and $s \approx 2$ across the SNR range we test. What produces a large exponent is *skew* — few dominant symbols against a heavy tail — which $\text{Ber}(\beta)$ at small β has and a smooth 256-ary posterior does not. Sharpening the latter only concentrates its mass and drives s back towards 2. In these settings it is the shape of the advice, more than the strength of the leakage, that decides whether Montanaro’s algorithm is worth deploying.

8 Conclusion

For product-distribution advice — the situation across a broad range of cryptographic settings — the Montanaro speedup exponent is not merely boundable but computable: exactly for finite-rank advice, and otherwise to any precision above the model’s atom-spacing and

floating-point floors, with a certificate each run assembles from its own arrays. The moments’ dependence on the surprisal law reduces the problem to one-dimensional convolutions; a single family of exponential tilts makes the computation provably accurate and numerically stable. The framework turns the asymptotic “there is a super-quadratic speedup” into a concrete, per-scheme “the guessing-moment exponent is s ”, with bounded discretization error, and at the finite parameters cryptology cares about and at settings where the entropy-based estimate of [7] is still loose.

Acknowledgements

The authors thank Alexander May, Franz J. Schreiber, Ronald de Wolf, Nicolai Kraus, Alexis Bagia, and Max Zhao for fruitful discussions and valuable feedback that helped bring this work to its final form. The authors used Anthropic’s Claude tools to assist with literature searches, numerical analyses, proofreading, and stylistic refinement. OpenAI’s GPT-5.6 Sol was additionally used to review the manuscript. All AI-generated output was critically evaluated by the authors, who take full responsibility for the content and conclusions of this work. The authors also gratefully acknowledge the following sources of funding. The group of C. Schubert and J. Seifert is supported by the DFG Priority Program SPP 2514, Berlin Quantum, as well as the BMFTR projects PQ-CCA and PraktiQOM. N. Paskarbit and M. Margraf are also supported by the BMFTR’s PQ-CCA project. M. J. Kramer acknowledges support from the DFG Priority Program SPP 2514; the BMFTR projects PraktiQOM, QuSol, and HYBRID++; the Munich Quantum Valley; Berlin Quantum; and the Clusters of Excellence MATH+ and ML4Q.

References

- [1] Martin R. Albrecht and Yixin Shen. 2023. Quantum Augmented Dual Attack. doi:10.48550/arXiv.2205.13983 arXiv:2205.13983 [quant-ph]. Also IACR ePrint 2022/656.
- [2] Michael Alekhnovich. 2011. More on Average Case vs Approximation Complexity. *computational complexity* 20, 4 (Dec. 2011), 755–786. doi:10.1007/s00037-011-0029-x
- [3] Noel T. Anderson, Jay-U. Chung, Shelby Kimmel, Da-Yeon Koh, and Xiaohan Ye. 2024. Improved Quantum Query Complexity on Easier Inputs. *Quantum* 8 (April 2024), 1309. doi:10.22331/q-2024-04-08-1309
- [4] Erdal Arıkan. 1996. An inequality on guessing and its application to sequential decoding. *IEEE Transactions on Information Theory* 42, 1 (Jan. 1996), 99–105. doi:10.1109/18.481781
- [5] R. R. Bahadur and R. Ranga Rao. 1960. On Deviations of the Sample Mean. *The Annals of Mathematical Statistics* 31, 4 (1960), 1015–1027. <https://www.jstor.org/stable/2237800>
- [6] Gustavo Banegas and Ricardo Villanueva-Polanco. 2025. On recovering block cipher secret keys in the cold boot attack setting. *Cryptography and Communications* 17, 2 (March 2025), 311–335. doi:10.1007/s12095-022-00625-z
- [7] Kaveh Bashiri, Timo Glaser, Alexander May, and Julian Nowakowski. 2026. Super-Quadratic Quantum Speed-ups and Guessing Many Likely Keys. In *Advances in Cryptology – EUROCRYPT 2026*. Springer Nature Switzerland, Cham, 457–480. doi:10.1007/978-3-032-25291-3_16
- [8] Avrim Blum, Adam Kalai, and Hal Wasserman. 2003. Noise-tolerant learning, the parity problem, and the statistical query model. *Journal of the ACM (JACM)* 50, 4 (July 2003), 506–519. doi:10.1145/792538.792543
- [9] Joppe Bos, Léo Ducas, Eike Kiltz, Tancrède Lepoint, Vadim Lyubashevsky, John M. Schanck, Peter Schwabe, Gregor Seiler, and Damien Stehlé. 2018. CRYSTALS - Kyber: A CCA-Secure Module-Lattice-Based KEM. In *2018 IEEE European Symposium on Security and Privacy (EuroS&P)*. 353–367. doi:10.1109/EuroSP.2018.00032
- [10] Alessandro Budroni and Erik Mårtensson. 2024. Further improvements of the estimation of key enumeration with applications to solving LWE. *Cryptography and Communications* 16, 5 (Sept. 2024), 1163–1182. doi:10.1007/s12095-024-00722-1
- [11] Suresh Chari, Josyula R. Rao, and Pankaj Rohatgi. 2003. Template Attacks. In *Cryptographic Hardware and Embedded Systems – CHES 2002*. Springer, Berlin, Heidelberg, 13–28. doi:10.1007/3-540-36400-5_3
- [12] Omar Choudary and Markus G. Kuhn. 2014. Efficient Template Attacks. In *Smart Card Research and Advanced Applications*. Springer International Publishing, Cham, 253–270. doi:10.1007/978-3-319-08302-5_17
- [13] Thomas M. Cover and Joy A. Thomas. 2006. *Elements of Information Theory* (2nd ed.). Wiley-Interscience, Hoboken, NJ. doi:10.1002/047174882X
- [14] Léo Ducas and Ludo N. Pulles. 2023. Does the Dual-Sieve Attack on Learning with Errors Even Work?. In *Advances in Cryptology – CRYPTO 2023*. Springer Nature Switzerland, Cham, 37–69. doi:10.1007/978-3-031-38548-3_2
- [15] Marc Fischlin and Evangelos Gkoumas. 2026. Bit Security of Quantum Key Search. In *Selected Areas in Cryptography – SAC 2025*. Springer Nature Switzerland, Cham, 264–290. doi:10.1007/978-3-032-10536-3_10
- [16] Marc Fischlin, Evangelos Gkoumas, and Gonne Kretschmer. 2026. Tighter Bit-Security Bounds in Quantum Key Search via the Chebyshev Distance. In *Post-Quantum Cryptography*. Springer Nature Switzerland, Cham, 71–96. doi:10.1007/978-3-032-22695-2_3
- [17] Valérie Gauthier-Umaña, Andrés M. Ochoa-Toro, and Ricardo Villanueva-Polanco. 2026. Seed Recovery From Probabilistic Leakage via Resource-Aware, Posterior-Guided Enumeration. *IEEE Access* 14 (2026), 17264–17277. doi:10.1109/ACCESS.2026.3659061
- [18] Lov K. Grover. 1996. A fast quantum mechanical algorithm for database search. In *Proceedings of the Twenty-Eighth Annual ACM Symposium on Theory of Computing*. Association for Computing Machinery, New York, NY, USA, 212–219. doi:10.1145/237814.237866
- [19] J. Alex Halderman, Seth D. Schoen, Nadia Heninger, William Clarkson, William Paul, Joseph A. Calandrino, Ariel J. Feldman, Jacob Appelbaum, and Edward W. Felten. 2009. Lest we remember: cold-boot attacks on encryption keys. *Commun. ACM* 52, 5 (May 2009), 91–98. doi:10.1145/1506409.1506429
- [20] Xiaoyu He, Xiaoming Sun, and Jialin Zhang. 2024. Quantum search with prior knowledge. *Science China Information Sciences* 67, 9 (Aug. 2024), 192503. doi:10.1007/s11432-023-3972-y Preliminary version arXiv:2009.08721 (2020).
- [21] Nicholas J. Higham. 2002. *Accuracy and Stability of Numerical Algorithms* (2nd ed.). Society for Industrial and Applied Mathematics, Philadelphia, PA. doi:10.1137/1.9780898718027
- [22] Jeffrey Hoffstein, Jill Pipher, and Joseph H. Silverman. 1998. NTRU: A ring-based public key cryptosystem. In *Algorithmic Number Theory*. Springer, Berlin, Heidelberg, 267–288. doi:10.1007/BFb0054868
- [23] Julien Maillard, Thomas Hiscock, Maxime Lecomte, and Christophe Clavier. 2024. Simulating SASCA on Keccak: Security Implications for Post-Quantum Cryptographic Schemes. In *21st International Conference on Security and Cryptography*. SCITEPRESS - Science and Technology Publications, Dijon, France, 518–527. doi:10.5220/0012787200003767
- [24] Daniel P. Martin, Ashley Montanaro, Elisabeth Oswald, and Dan Shepherd. 2018. Quantum Key Search with Side Channel Advice. In *Selected Areas in Cryptography – SAC 2017*. Springer International Publishing, Cham, 407–422. doi:10.1007/978-3-319-72565-9_21
- [25] James L. Massey. 1994. Guessing and entropy. In *Proceedings of 1994 IEEE International Symposium on Information Theory*. 204. doi:10.1109/ISIT.1994.394764
- [26] MATZOV. 2022. *Report on the Security of LWE: Improved Dual Lattice Attack*. Technical report. MATZOV, Israel Defense Forces. doi:10.5281/zenodo.6493704
- [27] Ashley Montanaro. 2011. Quantum Search with Advice. In *Theory of Quantum Computation, Communication, and Cryptography*. Springer, Berlin, Heidelberg, 77–93. doi:10.1007/978-3-642-18073-6_7
- [28] Valentin V. Petrov. 1975. *Sums of Independent Random Variables*. Springer, Berlin, Heidelberg. doi:10.1007/978-3-642-65809-9
- [29] Oded Regev. 2009. On lattices, learning with errors, random linear codes, and cryptography. *J. ACM* 56, 6 (Sept. 2009), 34:1–34:40. doi:10.1145/1568318.1568324 Preliminary version in STOC 2005.
- [30] Olivier Rioul. 2022. Variations on a Theme by Massey. *IEEE Transactions on Information Theory* 68, 5 (May 2022), 2813–2828. doi:10.1109/TIT.2022.3141264
- [31] Andrei Tănăsescu, Marios O. Choudary, Olivier Rioul, and Pantelimon George Popescu. 2021. Tight and Scalable Side-Channel Attack Evaluations through Asymptotically Optimal Massey-like Inequalities on Guessing Entropy. *Entropy* 23, 11 (Nov. 2021), 1538. doi:10.3390/e23111538
- [32] Nicolas Veyrat-Charvillat, Benoît Gérard, Mathieu Renaud, and François-Xavier Standaert. 2013. An Optimal Key Enumeration Algorithm and Its Application to Side-Channel Attacks. In *Selected Areas in Cryptography*. Springer, Berlin, Heidelberg, 390–406. doi:10.1007/978-3-642-35999-6_25

A The Arıkan scale of $f(\rho)$

Arıkan's inequality [4] brackets the optimal guessing moment $\mathbb{E}_P[G^\rho]$ directly: for the likelihood-ordered G and every $\rho > 0$,

$$(1 + \ln N)^{-\rho} \left(\sum_x P(x)^{\frac{1}{1+\rho}} \right)^{1+\rho} \leq \mathbb{E}_P[G^\rho] \leq \left(\sum_x P(x)^{\frac{1}{1+\rho}} \right)^{1+\rho},$$

and for a product source $\ln \left(\sum_x P^{1/(1+\rho)} \right)^{1+\rho} = \rho \sum_i H_{1/(1+\rho)}(p_i)$. (This is the natural-log original; the base-2 restatement in [7, Lem. 2.2] carries the slightly larger divisor $(1 + \log_2 |K|)^\rho$.) Since $\rho \ln(1 + \ln N) = O(\log m)$ for bounded alphabets, the bracket is tight to $O(\log m)$:

$$\begin{aligned} f(\rho) &= \ln \mathbb{E}_P[G^\rho] = \rho \sum_i H_{1/(1+\rho)}(p_i) + O(\log m) \\ &= \rho m \bar{H}_{1/(1+\rho)} \left(1 + O\left(\frac{\log m}{m}\right) \right). \end{aligned}$$

Hence $f(\frac{1}{2}) = \frac{1}{2} m \bar{H}_{2/3}(1 + o(1))$, $f(1) = m \bar{H}_{1/2}(1 + o(1))$, and $s = 2\bar{H}_{1/2}/\bar{H}_{2/3}(1 + o(1))$; this justifies the denominator scale in (7) and the recurrence of the orders $\frac{2}{3}$ and $\frac{1}{2}$.

B Proof of the local CLT

Reduction to integer summands, and the correct hypothesis. Dividing by η , the binned surprisals are $\tilde{s}_i(a) = \eta n_{ia}$ with $n_{ia} \in \mathbb{Z}$, so $\tilde{S}/\eta = \sum_i n_{i,X_i}$ is a sum of independent, integer-valued, non-identically distributed variables — a completely standard object, whose local CLT takes a hypothesis on the *product* of characteristic functions, frequency by frequency across the band $|t| \leq \pi/\eta$. A per-coordinate Cramér condition is the wrong shape here, and cannot hold: the band is sized to the span of the *sum*, while each coordinate is itself a lattice variable of coarser span $k_i\eta$, hence has $|\psi_i| = 1$ at its own resonance $2\pi/(k_i\eta)$ — strictly inside a band built for somebody else, and moving further inside as η is refined ($k_i = \text{gap}_i/\eta$ grows). The exponent Λ_η of (10) is exactly the product hypothesis, computed. The argument below is the classical Fourier route for lattice sums (see [28] for the framework); all constants are derived in place.

PROOF OF THE PRODUCT BOUND IN LEMMA 6.3. For coordinate i with centered binned values and tilted weights $p_{i,a}^{(\tau)}$,

$$|\psi_i(t)|^2 = \sum_{a,b} p_{i,a}^{(\tau)} p_{i,b}^{(\tau)} \cos(t(\tilde{s}_i(a) - \tilde{s}_i(b))) = 1 - X_i(t),$$

the imaginary parts cancelling in pairs, where $X_i(t)$ is defined as $\sum_{a < b} 4p_{i,a}^{(\tau)} p_{i,b}^{(\tau)} \sin^2(t\Delta_{iab}/2)$ by $1 - \cos \theta = 2\sin^2(\theta/2)$. Since $-\ln(1 - X) \geq X$ on $[0, 1)$, we have $-2 \sum_i \ln |\psi_i(t)| \geq \sum_i X_i(t)$, and hence $|\prod_i \psi_i(t)| \leq e^{-\Lambda(t)/2}$ with $\Lambda(t) := \sum_i X_i(t) \geq \Lambda_\eta$ on the band. We bound via the surrogate Λ rather than the exact $-\sum_i \ln |\psi_i|^2$ deliberately: the exact quantity is not a trigonometric polynomial, so its minimum is not FFT-certifiable, and its Lipschitz constant blows up where some $|\psi_i| \rightarrow 0$; Λ loses only where individual coordinates are strongly non-resonant — exactly where there is room to spare — and errs in the safe direction. $\square$

Certifying Λ_η in $\tilde{O}(m+1/\eta)$. With $4\sin^2(\pi u k) = 2(1 - \cos 2\pi u k)$ and $u := t\eta/2\pi$, $\Lambda(u) = W - \sum_{\ell \geq 1} c_\ell \cos(2\pi u k)$ is a cosine polynomial of degree $\ell_{\max} = \max_{i,a < b} k_{iab} = O(1/\eta)$. Herein, $W = \sum_i \sum_{a < b} 2p_{i,a}^{(\tau)} p_{i,b}^{(\tau)}$ and $c_\ell = \sum_{(i,a < b): k_{iab} = \ell} 2p_{i,a}^{(\tau)} p_{i,b}^{(\tau)}$. Accumulating

the c_ℓ costs $O(\sum_i b_i^2)$; one inverse FFT evaluates Λ on a uniform grid of N points in $O(N \log N)$; and $|\Lambda'(u)| \leq 2\pi \sum_\ell \ell c_\ell \leq 2\pi R/\eta$, so a grid of spacing h certifies $\min_u \Lambda \geq \min_{\text{grid}} \Lambda - 2\pi R h/\eta$. Relative accuracy ε on $\Lambda = \Theta(m)$ with $R = \Theta(m)$ needs $h = \Theta(\varepsilon\eta)$, i.e. $N = O(1/(\varepsilon\eta))$: total $\tilde{O}(m+1/\eta)$, below the engine's own cost. In practice the candidate minima are the resonances $u = j/k_{iab}$; evaluate those exactly and let the grid certify that nothing lower sits between them.

PROOF OF LEMMA 6.3. Let $\psi = \prod_i \psi_i$. The span- η inversion formula and the Gaussian inversion are

$$\begin{aligned} p_k &= \frac{\eta}{2\pi} \int_{-\pi/\eta}^{\pi/\eta} e^{-its_k} \psi(t) dt, \\ \frac{1}{\sigma_\tau} \varphi(s_k/\sigma_\tau) &= \frac{1}{2\pi} \int_{\mathbb{R}} e^{-its_k} e^{-\sigma_\tau^2 t^2/2} dt. \end{aligned}$$

Subtracting the two formulas and using $|e^{-its_k}| = 1$,

$$\left| g_k - \frac{1}{\sigma_\tau} \varphi(s_k/\sigma_\tau) \right| \leq \frac{J_1 + J_2 + J_3}{2\pi},$$

where $J_1 = \int_{|t| \leq \delta_1} |\psi - e^{-\sigma_\tau^2 t^2/2}| dt$ collects the low frequencies, $J_2 = \int_{\delta_1 < |t| \leq \pi/\eta} |\psi| dt$ the high frequencies, and $J_3 = \int_{|t| > \delta_1} e^{-\sigma_\tau^2 t^2/2} dt$ the Gaussian tail, with the split point δ_1 of Lemma 6.3, whose two arms are exactly the Taylor radius and the branch radius of the logarithm below.

Low frequency J_1 (Edgeworth, with the quartic term). Write $w_i(t) := \psi_i(t) - 1$, so that $|w_i| \leq \sigma_i^2 t^2/2$ and $|\psi_i - 1 + \sigma_i^2 t^2/2| \leq \beta_i |t|^3/6$ (third-order Taylor of the characteristic function). The second arm of δ_1 gives $\max_i |w_i(t)| \leq \max_i \sigma_i^2 t^2/2 \leq \frac{1}{2}$ on $|t| \leq \delta_1$ — the branch condition, enforced by construction — so the principal logarithm satisfies $|\ln \psi_i - w_i| \leq |w_i|^2$. Summing over coordinates,

$$\ln \psi(t) = -\frac{\sigma_\tau^2 t^2}{2} + R, \quad |R| \leq \frac{\beta_\tau |t|^3}{6} + Q(t), \quad Q(t) := \sum_i |w_i(t)|^2.$$

The cubic-only bound $|R| \leq \beta_\tau |t|^3/6$ that a summed “Taylor with remainder” would suggest is *false* in general — it conflates the expansion of ψ_i with that of $\ln \psi_i$, and the measured ratio $|R|/(\beta_\tau |t|^3/6)$ reaches 1.09 on i.i.d. Bernoulli(0.8) advice at $m = 256$ — so the quartic term Q is carried, not hedged. The choice of δ_1 forces $\beta_\tau |t|^3/6 \leq \sigma_\tau^2 t^2/4$, whence $|e^R - 1| \leq |R|e^{|R|}$ gives

$$|\psi - e^{-\sigma_\tau^2 t^2/2}| \leq \left(\frac{\beta_\tau |t|^3}{6} + Q(t) \right) e^{Q(t)} e^{-\sigma_\tau^2 t^2/4}.$$

Hence $\frac{\sigma_\tau}{2\pi} J_1 \leq \frac{4}{3\pi} \frac{\beta_\tau}{\sigma_\tau^3} + q_\tau$ with the *quartic correction*

$$q_\tau := \frac{\sigma_\tau}{2\pi} \int_{|t| \leq \delta_1} \left[\frac{\beta_\tau |t|^3}{6} (e^{Q(t)} - 1) + Q(t) e^{Q(t)} \right] e^{-\sigma_\tau^2 t^2/4} dt,$$

a computed integral: Q is an explicit trigonometric expression in the marginals. Since $|w_i| \leq \sigma_i^2 t^2/2$ pointwise, $q_\tau = O((\max_i \sigma_i^2)/\sigma_\tau^2)$, which is $O(1/m)$ under the regularity of Corollary 6.8 — asymptotically below the cubic term, but numerically 14–38% of it at $m = 256$ across representative ensembles, so an explicit ε_τ must include it.

High frequency J_2 (lattice span). By the product bound, $|\psi(t)| \leq e^{-\Lambda_\eta/2}$ on $\delta_1 \leq |t| \leq \pi/\eta$, so $J_2 \leq \frac{2\pi}{\eta} e^{-\Lambda_\eta/2}$ and $\frac{\sigma_\tau}{2\pi} J_2 \leq \frac{\sigma_\tau}{\eta} e^{-\Lambda_\eta/2}$ — a computed number, not an order symbol. Whether it is small is precisely the content of the hypothesis $\Lambda_\eta \geq \lambda m$; that neither gcd = 1 nor bounded per-coordinate leakage implies this is shown by the counterexample in Remark 6.2.

Gaussian tail J_3 . This is the Gaussian's own integral beyond δ_1 , *not* aliasing: the genuine lattice images beyond $\pm\pi/\eta$ are already excluded by truncating the inversion integral to one Nyquist period. Exactly, $J_3 = \frac{2\sqrt{2\pi}}{\sigma_\tau} \Phi(\sigma_\tau \delta_1)$, so $\frac{\sigma_\tau}{2\pi} J_3 = \sqrt{2/\pi} \Phi(\sigma_\tau \delta_1)$ – evaluated, not asserted; with the computed δ_1 it is below 10^{-15} on the tested ensembles at $m = 256$, and it is reported rather than absorbed.

Assembling the three bounds gives the display of Lemma 6.3. $\square$

Why the local CLT cannot be dropped for large deviations. The quantity the accuracy analysis needs is $\ln[\tilde{F}(t + m\eta)/\tilde{F}(t)]$ for an increment $m\eta = O(\sqrt{m})$ in the scaling regime – *sublinear* on the large-deviation scale. Cramér-level theory controls $\ln \tilde{F}$ only to $o(m)$, larger than the entire increment; resolving it requires sharp large deviations of Bahadur–Rao type, and those rest on exactly the density-level control proved here (cf. Remark C.1, the same point from the other direction).

Remark B.1 (connection to the Fourier sup-density). $\sup_k g_k \leq M_\tau := \frac{1}{2\pi} \int |\psi|$ by the triangle inequality on the inversion formula; up to the Gaussian tail J_3 , the high-frequency term of ε_τ is the $|\tau| > \delta_1$ part of M_τ rescaled by σ_τ , while the low-frequency mass reproduces the Gaussian peak $1/(\sqrt{2\pi}\sigma_\tau)$.

C Proofs of the rate lemma and the edge term

We first prove the value-level reversed-hazard bound (Lemma 6.6), then the sharp multiplicative equality (Lemma 6.4); the section closes with the explicit propagation into ε_m and the regularity micro-lemma behind Corollary 6.8.

PROOF OF LEMMA 6.6. Fix $t \geq H$ in the saddle range and write $\tau < 0$ for the binned saddle at t ; it differs from τ_t by the computed shift of the comparison paragraph below, absorbed into ε_r . Put $z = |\tau|\sigma_\tau$. Tilting the binned count law by τ and solving the tilt for $\tilde{v}$, level by level, gives the exact identity

$$\frac{\tilde{v}(t)}{\tilde{F}(t)} = \frac{\mathbb{P}_\tau[\tilde{S} = t]}{\sum_{j \geq 0} e^{\tau j \eta} \mathbb{P}_\tau[\tilde{S} = t - j\eta]}.$$

Lemma 6.3 controls every point mass appearing here: $\mathbb{P}_\tau[\tilde{S} = y] = \frac{\eta}{\sigma_\tau} (\varphi(\tilde{y}/\sigma_\tau) + r(y))$ with $|r| \leq \varepsilon_r$ and $\tilde{y}$ the centered argument. At the saddle the anchor sits at the tilted mean, so across the displacements that matter the Gaussian factor satisfies $\varphi(\tilde{y}/\sigma_\tau) \geq \varphi(0)(1 - O(z^{-2}))$, and the point-mass control is a *relative* $\sqrt{2\pi}\varepsilon_\tau(1 + o(1))$ there – the $\sqrt{2\pi}$ being $1/\varphi(0)$. In the denominator the weight $e^{\tau j \eta}$ confines the sum to displacements $j\eta = O(1/|\tau|)$, across which the Gaussian factor moves by $1 - O(z^{-2})$; hence

$$\sum_{j \geq 0} e^{\tau j \eta} \mathbb{P}_\tau[\tilde{S} = t - j\eta] = \frac{\eta \varphi(\tilde{t}/\sigma_\tau)/\sigma_\tau}{1 - e^{-|\tau|\eta}} (1 + O(\sqrt{2\pi}\varepsilon_\tau) + O(z^{-2})),$$

and the ratio obeys

$$\frac{\tilde{v}(t)}{\tilde{F}(t)} \leq (1 - e^{-|\tau|\eta}) (1 + 2\sqrt{2\pi}\varepsilon_\tau(1 + O(\varepsilon_\tau)) + O(z^{-2})) \leq |\tau|\eta (1 + \varepsilon_r(t)),$$

using the exact $1 - e^{-u} \leq u$; the $O(|\tau|\eta)$ term of ε_r absorbs the second-order geometric correction and the saddle shift. Everything here is a value statement: nothing is differentiated, the geometric factor is exact, and the correction carries z^{-2} itself, so no implicit $z \gg 1$ enters and the near-uniform regime is covered.

Coverage of $t \geq H$ outside the saddle range. For $z_t = O(1) - t$ approaching and beyond μ_U , where Lemma 6.4's hypothesis lapses – write the single-step ratio through the Gaussian hazard $\lambda = \varphi/\Phi$: its model value is $(\eta/\sigma_t) \lambda(z_t) (1 + O(\varepsilon_r))$, and $\lambda(z_t) = O(1)$ there, so the ratio is $O(\eta/\sigma_t)$ – an order below the $|\tau_t|\eta$ of the saddle range. The supremum over $\{t \geq H\}$ is therefore governed by the saddle range, where the exact $|\tau_t| \leq 1$ of Corollary 6.5 closes the statement. $\square$

For the binned chain, $\tilde{\Lambda}_U(\tau) = \ln \mathbb{E}_U[e^{\tau \tilde{S}}]$ satisfies $|\tilde{\Lambda}_U(\tau) - \Lambda_U(\tau)| \leq |\tau|m\eta$ (since $0 \leq \tilde{S} - S < m\eta$); for the derivative, the change-of-measure contribution is $O(|\tau|\sigma_\tau \text{sd}(\tilde{S} - S))$ with $\text{sd}(\tilde{S} - S) = O(\eta\sqrt{m})$, so dividing by $\Lambda'_U = \sigma_\tau^2$ the saddles obey $|\tilde{\tau}_t - \tau_t| = O(m\eta/\sigma_\tau^2 + \eta\sqrt{m}/\sigma_\tau)$, a computed scalar – $O(\eta)$ under Corollary 6.8 – absorbed into ε_r below.

PROOF OF LEMMA 6.4. Write $\tau = \tau_t < 0$, $\sigma = \sigma_\tau$, $z = |\tau|\sigma$, $u = h/\sigma$, $h = m\eta$. We prove the one-sided multiplicative statement for the binned $\tilde{F}$; the same proof applies verbatim to the true F_S . By the exact tilting identity (9), $\ln[\tilde{F}(t + h)/\tilde{F}(t)] = \ln[M(h)/M(0)]$. Split $M = \mathcal{G} + \mathcal{R}$ into Gaussian surrogate and local-CLT residual,

$$\mathcal{G}(a) = \int_{-\infty}^a e^{|\tau|s} \frac{1}{\sigma} \varphi(s/\sigma) ds, \\ \mathcal{R}(a) = \sum_{s_k \leq a} e^{|\tau|s_k} r(s_k) \eta, \quad |r| \leq \varepsilon_r/\sigma;$$

the lattice-sum-versus-integral mismatch on the smooth part is itself $O(\varepsilon_r)$ (Euler–Maclaurin) and is absorbed into r . Completing the square,

$$\mathcal{G}(a) = e^{z^2/2} \tilde{\Phi}\left(z - \frac{a}{\sigma}\right), \quad (14)$$

and Mills' bounds give $\mathcal{G}(a) = \frac{e^{|\tau|a}}{z} \mu_a$ with $\mu_a = \Theta_{u_0}(1)$ uniformly for $0 \leq a \leq h$ (only boundedness is needed here; the sharp expansion belongs to the *ratio* $\mathcal{G}(h)/\mathcal{G}(0)$, derived in Part A). The residual has the *same* envelope:

$$\left| \int_{-\infty}^a e^{|\tau|s} r(s) ds \right| \leq \frac{\varepsilon_r}{\sigma} \cdot \frac{e^{|\tau|a}}{|\tau|} = \frac{\varepsilon_r e^{|\tau|a}}{z}, \quad (15) \\ \text{so } M(a) = \mathcal{G}(a)(1 + \theta_a), \quad |\theta_a| \leq c_1 \varepsilon_r.$$

Here and below, c_1 denotes an absolute constant.

Part A (Gaussian part). With $u = h/\sigma$,

$$A := \ln \frac{\mathcal{G}(h)}{\mathcal{G}(0)} = \ln \tilde{\Phi}(z - u) - \ln \tilde{\Phi}(z) = \int_{z-u}^z \lambda(x) dx,$$

with $\lambda := \varphi/\Phi$, using $\frac{d}{dx} \ln \tilde{\Phi}(x) = -\lambda(x)$. The Gaussian hazard rate expands as $\lambda(x) = x + \frac{1}{x} + O(x^{-3})$, so $A = (zu - \frac{u^2}{2}) + \ln \frac{z}{z-u} + O(uz^{-3})$. Now $zu = |\tau|h$ and $\ln \frac{z}{z-u} = \frac{u}{z} + \frac{u^2}{2z^2} + O(u^3/z^3)$; dividing by $|\tau|h = zu$,

$$A = |\tau|h \left(1 - \frac{u}{2z} + \frac{1}{z^2} + O\left(\frac{u}{z^3} + \frac{1}{z^4}\right) \right) = |\tau|h (1 + O(u/z) + O(z^{-2})), \quad (16)$$

valid for $u \leq u_0$; under Corollary 6.8, $\frac{u}{z} = \Theta(\eta)$ and $\frac{1}{z^2} = \Theta(1/m)$. Differencing $\ln \tilde{\Phi}$ *before* expanding is what removes the additive floor: a two-sided estimate would leave a standalone Mills correction $+2/z^2$, whereas here it survives only inside the relative factor, as the $+1/z^2$ Bahadur–Rao correction of Lemma 6.6.

Part B (residual, by discrete difference). Put $\theta_a := \mathcal{R}(a)/\mathcal{G}(a)$, so $\ln[M(h)/M(0)] = A + B$ with $B := \ln \frac{1+\theta_h}{1+\theta_0}$; the per-point bound

(15) gives $|\theta_a| \leq c_1 \varepsilon_\tau$. The improvement comes from bounding the variation across the window — but $\mathcal{R}$ is a lattice sum, so $a \mapsto \theta_a$ is a step-drift function, not differentiable, and its variation has two sources, bounded separately. (i) *Jumps*. At a lattice point s_k in the window, θ jumps by $|\Delta\theta| \leq e^{|\tau|s_k} |r(s_k)| \eta / \mathcal{G}(s_k)$; with $\mathcal{G}(s_k) = e^{|\tau|s_k} \mu_{s_k} / z$, $\mu_{s_k} = \Theta_{u_0}(1)$ and $|r| \leq \varepsilon_\tau / \sigma$, this is $|\Delta\theta| \leq c_1 |\tau| \varepsilon_\tau \eta$; summing over the $h/\eta = m$ lattice points in the window contributes $\leq c_1 |\tau| h \varepsilon_\tau$ in total.

(ii) *Drift*. Between lattice points we have $\theta'_a = -\theta_a \mathcal{G}'(a) / \mathcal{G}(a)$ because $\mathcal{R}$ is constant — a legitimate differentiation of the explicit Gaussian integral $\mathcal{G}$, not of an approximation carrying a remainder. Also $\mathcal{G}' / \mathcal{G} = |\tau| (1 + O(u/z) + O(z^{-2}))$ with $|\theta_a| \leq c_1 \varepsilon_\tau$, so the drift across the window contributes $\leq c_1 |\tau| h \varepsilon_\tau$. Adding the two, we get

$$|\theta_h - \theta_0| \leq c_1 |\tau| h \varepsilon_\tau = |\tau| h \cdot O(\varepsilon_\tau), \quad (17)$$

so $|B| = |\tau| h \cdot O(\varepsilon_\tau)$; the bound is window-proportional because $\mathcal{R}$ and $\mathcal{G}$ share the envelope $e^{|\tau|a} / z$. Combining, $\ln[M(h)/M(0)] = A + B = |\tau| m \eta (1 + O(u/z) + O(z^{-2}) + O(\varepsilon_\tau))$. $\square$

Remark C.1 (CDF-level Berry–Esseen is insufficient). Integrating (9) by parts against the tilted CDF and using a uniform Berry–Esseen bound $\text{BE}_\tau = O(m^{-1/2})$ produces a boundary term $e^{|\tau|a} (F_S^\tau - \Phi)(a)$ of size $e^{|\tau|a} \text{BE}_\tau$, whereas the answer is $\mathcal{G}(a) \approx e^{|\tau|a} / (z\sqrt{2\pi})$; their ratio is $z \text{BE}_\tau = \Theta(1)$. The CDF error is the same order as the answer, so the density-level statement is genuinely required — and it is supplied by the *lattice* Lemma 6.3 applied to $\tilde{F}$.

Propagation into ε_m , explicit. Collecting the chain behind Theorem 6.7: Lemma 6.6 caps each single-step hazard by $q = \eta(1 + \varepsilon_r)$, so telescoping gives $\Delta(x) \leq m\eta/(1 - q)$ on $\{S \geq H\}$ with $\varepsilon_r \leq 2\sqrt{2\pi} \varepsilon_\tau (1 + O(\varepsilon_\tau)) + \bar{z}^{-2} + O(\eta)$, where $\bar{z}^{-2} := \sup\{z^{-2} : t \geq H, |\tau_t| \geq \tau_0\}$ runs over the saddle range only — the bulk $|\tau_t| < \tau_0$ is covered by Lemma 6.6's $O(\eta/\sigma_t)$ estimate, and at the edge $z_H^{-2} = 1/(m\text{Var}_1)$ — the localization adds an exponentially small remainder, and the exact ratio identity divides by $f_\eta(\frac{1}{2})$. Hence $\varepsilon_m \leq 2\sqrt{2\pi} \varepsilon_\tau (1 + O(\varepsilon_\tau)) + \bar{z}^{-2} + O(\eta) + O(\log(1 + \ln N)/f_\eta(\frac{1}{2})) + O(e^{-\Theta(m)})$, with ε_τ the computed constant of Lemma 6.3 and $\bar{z}^{-2}$ read off the computed saddle scalars. This display is what makes ε_m genuinely evaluable: an explicit ε_τ without the propagation constant, or the removal of $\bar{z}^{-2}$ from the boxed bound without either, would each leave ε_m less explicit than before. Numerically the first term dominates: at $m = 256$ on heterogeneous Bernoulli advice, $\varepsilon_\tau \approx 0.070$ against $1/(m\text{Var}_1) \approx 0.013$.

The regularity micro-lemma behind Corollary 6.8. Write $r_i := \max_a s_i(a) - \min_a s_i(a) \leq c$, with bounded alphabets, and let Y_i be the tilted per-coordinate surprisals, $\tau \in [-1, 0]$. *Upper:* $\text{Var}_\tau(Y_i) \leq r_i^2/4$, so $\sigma_\tau^2 \leq \frac{1}{4} \sum_i r_i^2 \leq \frac{1}{4} cR = O(m)$. *Lower:* for $|\tau| \leq 1$ the tilted weights satisfy $p_{i,a}^{(\tau)} \geq e^{-c} \min_a p_{i,a}$, bounded away from 0 and 1 by constants depending only on c and the alphabet sizes, hence $\text{Var}_\tau(Y_i) \geq c' r_i^2$; Cauchy–Schwarz gives $\sum_i r_i^2 \geq R^2/m = \Theta(m)$, so $\sigma_\tau^2 = \Theta(m)$. *Third moment:* $|Y_i - \mathbb{E}Y_i| \leq r_i \leq c$, so $\mathbb{E}|Y_i - \mathbb{E}Y_i|^3 \leq c \text{Var}_\tau(Y_i)$, whence $\beta_\tau \leq c \sigma_\tau^2$ and $\beta_\tau/\sigma_\tau^3 \leq c/\sigma_\tau = O(m^{-1/2})$, uniformly over the tilt range. Note what is *not* needed: any per-coordinate variance floor — coordinates with $r_i = 0$ contribute nothing to either sum and are harmless. Substituting into Lemma 6.3 and the propagation display above yields Corollary 6.8. $\square$

C.1 Proof of the scaling law

PROOF OF THEOREM 6.7. Split $\mathbb{E}_P[\tilde{G}^\rho]$ at the level H : above it the rate is controlled (Lemma 6.6), below it nothing needs to be controlled.

On $\{S \geq H\}$. By Lemma 6.1, $G^\rho e^{-\rho\Delta} \leq \tilde{\mathcal{A}}_\rho(\tilde{S}) \leq G^\rho e^{\rho\Delta}$ pointwise. Telescope in steps of η over the window's lattice points $t_k \geq H$ and put $h_k := \tilde{v}(t_k)/\tilde{F}(t_k)$. Since $\tilde{F}(t_{k-1}) = \tilde{F}(t_k) - \tilde{v}(t_k)$, each increment is $\ln[\tilde{F}(t_k)/\tilde{F}(t_{k-1})] = -\ln(1 - h_k) \leq h_k/(1 - h_k)$, which exceeds h_k : the inclusive hazard may not be summed directly. By Lemma 6.6 each h_k is at most $|\tau_{t_k}| \eta (1 + \varepsilon_r)$, and the exact convexity bound $|\tau_t| \leq \theta^* = 1$ of Corollary 6.5 on $[H, \mu_U]$, with the bulk beyond covered by Lemma 6.6, caps them all by $q := \eta(1 + \varepsilon_r) < 1$, giving $\Delta(x) \leq m\eta/(1 - q) = (1 + \varepsilon_m)m\eta$ on $\{S \geq H\}$ with $1 + \varepsilon_m = (1 + \varepsilon_r)/(1 - q)$. Only the pointwise rate enters; the sharp equality of Lemma 6.4 is a stronger statement that this route does not need. Hence

$$e^{-\rho(1+\varepsilon_m)m\eta} \mathbb{E}_P[G^\rho; S \geq H] \leq \mathbb{E}_P[\tilde{\mathcal{A}}_\rho(\tilde{S}); S \geq H] \leq e^{\rho(1+\varepsilon_m)m\eta} e^{f(\rho)}.$$

Localization: $\{S < H\}$ carries no weight. The restriction to $\{S \geq H\}$ is essential — below H the rate $|\tau_t|$ blows up and Δ with it — but the region contributes only exponentially little to either moment: guessing moments are dominated by *harder-than-typical* keys (the Rényi level $\bar{H}_{1/(1+\rho)} > \bar{H}_1$), while every key below the entropy level has rank at most e^H . By (4) and $\tilde{F} \leq F_S$ (8), $\tilde{\mathcal{A}}_\rho(\tilde{S}(x)) \leq (N \cdot \tilde{F}(\tilde{S}(x)))^\rho \leq e^{\rho\tilde{S}(x)} \leq e^{\rho(S(x)+m\eta)}$ and $G(x) \leq G^*(x) \leq e^{S(x)}$, so on $\{S < H\}$,

$$\mathbb{E}_P[\tilde{\mathcal{A}}_\rho(\tilde{S}); S < H] \leq e^{\rho(H+m\eta)}, \quad \mathbb{E}_P[G^\rho; S < H] \leq e^{\rho H}.$$

Meanwhile, by (1),

$$f(\rho) - \rho H = \rho m (\bar{H}_{1/(1+\rho)} - \bar{H}_1) + O(\log(1 + \ln N)),$$

which, by strict monotonicity of $\alpha \mapsto H_\alpha$ and $1/(1 + \rho) < 1$, is extensive whenever the advice is non-uniform on a constant fraction of coordinates (otherwise $s \rightarrow 2$ and the exponent ceases to be a meaningful target). Combining the two regions, $|f_\eta(\rho) - f(\rho)| \leq \rho m \eta (1 + \varepsilon_m) + O(e^{-\Theta(m)})$, with the exponentially small remainder absorbed into ε_m .

Propagation — exact. Write $f_\eta(\rho) = f(\rho) + e_\rho$ with $|e_\rho| \leq \rho m \eta (1 + \varepsilon_m)$. Since $f(1) = s f(\frac{1}{2})$ by the definition of s ,

$$s_\eta - s = \frac{f(1) + e_1}{f(\frac{1}{2}) + e_{1/2}} - \frac{f(1)}{f(\frac{1}{2})} = \frac{e_1 - s e_{1/2}}{f_\eta(\frac{1}{2})} \quad \text{exactly, no remainder,}$$

so $|s_\eta - s| \leq \frac{m\eta(1+s/2)}{f_\eta(\frac{1}{2})} (1 + \varepsilon_m)$; substituting the asymptotic scale

$f_\eta(\frac{1}{2}) = \frac{1}{2} m \bar{H}_{2/3} (1 + o(1))$ of (1) gives the display. The identity is worth noting in its own right: the certificate route divides by the *computed* $f_\eta(\frac{1}{2})$ instead of the asymptotic scale, and thereby avoids (1) altogether (Proposition 6.9). $\square$

D Proofs for the computable certificate

PROOF OF THE BRACKET (12). Strict rounding gives $0 < \tilde{S} - S \leq m\eta$ pointwise. *Localization within a bin.* If $\tilde{S}(x) = t$ then $S(x) < t$ by strictness and $S(x) \geq \tilde{S}(x) - m\eta = t - m\eta$, so $S(x) \in [t - m\eta, t)$. Since $\tilde{F}$ is a right-continuous step function on $\eta\mathbb{Z}$ and $m\eta \in \eta\mathbb{Z}$, the map $u \mapsto \ln[\tilde{F}(u + m\eta)/\tilde{F}(u)]$ is constant on each cell $[\ell, \ell + \eta)$, and

the cells meeting $[t - m\eta, t)$ are indexed by $\ell \in \{t - m\eta, \dots, t - \eta\}$; hence

$$\Delta(x) = \ln \frac{\tilde{F}(S(x) + m\eta)}{\tilde{F}(S(x))} \leq D(t) \quad \text{whenever } \tilde{S}(x) = t. \quad (18)$$

Three bounds on the rank of a key in bin t . Let $\tilde{S}(x) = t$. (1) *Rank window*: the endpoint comparisons of Lemma 6.1 give $N \cdot \tilde{F}(S(x)) < G(x) \leq N \cdot \tilde{F}(S(x) + m\eta)$; with $S(x) \in [t - m\eta, t)$ and $\tilde{F}$ non-decreasing, $N \cdot \tilde{F}(t - m\eta) < G(x) \leq N \cdot \tilde{F}(t + m\eta)$, and G being an integer also $G(x) \geq 1$. (2) *Chernoff cap*: $G(x) \leq N \cdot F_S(S(x)) \leq e^{S(x)} < e^t$ by (4) – exact, with no distributional input. (3) *Window ratio*: if $D(t) < \infty$, Lemma 6.1 places $G(x)^\rho$ and $\tilde{\mathcal{A}}_\rho(\tilde{S}(x)) = \tilde{\mathcal{A}}_\rho(t)$ within $e^{\pm \rho \Delta(x)}$ of each other, and $\Delta(x) \leq D(t)$ by (18). Combining, $G(x)^\rho \in [\text{lo}_\rho(t), \text{hi}_\rho(t)]$ for every x in bin t ; multiplying by $P(x)$, summing over the bin – whose total P -mass is $\tilde{\pi}(t)$ – and then over t gives (12). No threshold, no tail term, no localization argument: every level is handled by whichever of the three bounds is tightest there. $\square$

PROOF OF THE CERTIFICATE (13). From (12), $f(\rho)$ lies in a computed interval, and $E_\rho := \max \{f_\eta(\rho) - \ln \sum_t \tilde{\pi} \text{lo}_\rho, \ln \sum_t \tilde{\pi} \text{hi}_\rho - f_\eta(\rho)\} \geq |e_\rho|$. By the exact ratio identity in the proof of Theorem 6.7, $s_\eta - s = (e_1 - s e_{1/2})/f_\eta(\frac{1}{2})$. Write $x := |s_\eta - s|$; since $s \leq s_\eta + x$ and $s > 0$,

$$x \leq \frac{E_1 + s E_{1/2}}{f_\eta(\frac{1}{2})} \leq \frac{E_1 + (s_\eta + x) E_{1/2}}{f_\eta(\frac{1}{2})},$$

so $x (f_\eta(\frac{1}{2}) - E_{1/2}) \leq E_1 + s_\eta E_{1/2}$, and (13) follows whenever $f_\eta(\frac{1}{2}) > E_{1/2}$. The denominator $f_\eta(\frac{1}{2}) - E_{1/2} \leq f(\frac{1}{2})$ is exactly the lower bound on the true denominator that an a-priori route would need Arkan to supply – obtained here by *computing* $f_\eta(\frac{1}{2})$, which is why no Arkan input, and hence no asymptotic ingredient, appears. $\square$

Which hypotheses the certificate uses. The inputs to the two proofs are: product structure (so the engine exists and $\tilde{S}$ is a sum of per-coordinate roundings); the positivity convention; strict upward rounding (a definition); Lemma 6.1, proved by counting; the exact (4); and arithmetic. The regime constant ε_0 is never used – no expansion is taken – and no local CLT is invoked. *Validity is not usefulness*: the certificate stays valid where the scaling law fails, but it is not guaranteed to be informative – at Bernoulli(0.999) advice, $m = 256$, it returns 15.3: rigorous and useless. The correct formulation is *valid under the engine assumptions alone; informative exactly where the level structure is rich enough to measure*. In the families where it is vacuous the level count is polynomial, so the finite-rank exact engine covers them (Section 4.1); and where the scaling law’s hypotheses fail – the near-commensurate ensembles of Remark 6.2 – the certificate degrades *visibly*, by returning a larger number, rather than silently ceasing to hold.

E Proofs of the engine lemmas

E.1 Optimality and the level averages

PROOF OF LEMMA 3.1(i). $\mathbb{E}_P[G^\rho] = \sum_{r=1}^N r^\rho P(G^{-1}(r))$ (reindexed by rank) depends on G only through the sequence $(P(G^{-1}(r)))_r$, which is a permutation of the fixed multiset $\{P(x)\}$. Since $(r^\rho)_r$ is

strictly increasing, the rearrangement inequality (Hardy–Littlewood–Pólya) shows the pairing is minimized exactly when $r \mapsto P(G^{-1}(r))$ is non-increasing, i.e. when G is likelihood-ordered; every likelihood-ordered G produces the *same* non-increasing sequence (within-block permutations do not change it), so the minimum is attained and shared. Note the minimizer does not depend on ρ : one and the same enumeration is optimal for the classical ($\rho = 1$) and the Montanaro ($\rho = \frac{1}{2}$) moment. $\square$

PROOF OF THE BRACKET (2). Write $b = N \cdot F_S(t)$, $a = N \cdot F_S(t^-)$, $n = b - a \geq 1$. *Upper*: every $r \in J_t$ has $r \leq b$, so $\mathcal{A}_\rho(t) \leq b^\rho$, with equality iff $J_t = \{b\}$, i.e. $n = 1$. *Lower*: $x \mapsto x^\rho$ is increasing, so the block sum is a right Riemann sum, $\sum_{r=a+1}^b r^\rho \geq \int_a^b x^\rho dx = (b^{\rho+1} - a^{\rho+1})/(\rho+1)$; with the occupancy ratio $\beta := n/b \in (0, 1]$,

$$\mathcal{A}_\rho(t) \geq b^\rho \frac{1 - (1 - \beta)^{\rho+1}}{(\rho+1)\beta} \geq \frac{b^\rho}{1 + \rho},$$

the last step because $\beta \mapsto 1 - (1 - \beta)^c$ with $c \geq 1$ is concave, vanishes at $\beta = 0$ and equals 1 at $\beta = 1$, hence dominates the chord β . The middle expression shows $\mathcal{A}_\rho(t) = b^\rho (1 + O(\beta))$: the block-maximal surrogate is the thin-level limit, and $\beta = v(t)/F_S(t)$ is precisely the discrete reversed-hazard ratio studied by Lemma 6.6. $\square$

Certified evaluation of $\mathcal{A}_\rho$. For $\rho = 1$ the block sum is closed-form, $\sum_{r=a+1}^b r = [b(b+1) - a(a+1)]/2$. For $\rho = \frac{1}{2}$, since $\sqrt{\cdot}$ is increasing and concave, the midpoint rule (tangent above the graph, so $\int_{r-1/2}^{r+1/2} f \leq f(r)$) and the trapezoid rule (chord below it, so $\int_{r-1}^r f \geq [f(r-1) + f(r)]/2$), summed and telescoped over the block, give the elementary two-sided closed-form bracket

$$\frac{2}{3} \left[(b + \frac{1}{2})^{3/2} - (a + \frac{1}{2})^{3/2} \right] \leq \sum_{r=a+1}^b \sqrt{r} \leq \frac{2}{3} \left[b^{3/2} - a^{3/2} \right] + \frac{\sqrt{b} - \sqrt{a}}{2},$$

of relative width $O(B^{-3/2})$ in the block size $B = n$. The evaluation rule at target relative precision ε : blocks of $B \leq B_\star := \varepsilon^{-2/3}$ are summed directly, the ranks being integer-exact in IEEE double for $\varepsilon \geq u_m$ and the accumulation sitting inside Lemma 4.4’s budget, and larger blocks use the bracket, whose relative width is then at most ε , within the u_m poly budget of Lemma 4.4; per-level cost is $O(B_\star)$, a constant in m . (In practice the crossover is immaterial: levels are either thin – heterogeneous advice, generically $n(t) = O(1)$ – or astronomically above $B_\star$, as for i.i.d. advice.) Carrying the bracket as an interval instead makes the evaluation certified by construction. Numerically, the ratio $N^{-\rho} \mathcal{A}_\rho(t)$ is formed in the occupancy ratio β via $\text{expm1}/\log1p$ – e.g. $b^{3/2} - a^{3/2} = b^{3/2} \cdot (-\text{expm1}(\frac{3}{2} \log1p(-\beta)))$ – never as a difference of close large powers, which would lose $\approx \log_{10}(1/\beta)$ digits on thin levels, the generic case for heterogeneous advice; in this form the evaluation is accurate to a few ulps uniformly in β and is covered by the error budget of Lemma 4.4.

Remark E.1 (cost of the block-maximal shortcut). Evaluating the moments at the block-maximal rank – $(N \cdot F_S(t))^\rho$ in place of $\mathcal{A}_\rho(t)$ – is tempting, and (2) prices it: the surrogate exponents satisfy $0 \leq f^\square(\rho) - f(\rho) \leq \ln(1 + \rho)$, and the exact ratio identity in the proof of Theorem 6.7 turns this into

$$|s^\square - s| \leq \frac{\ln 2 + s \ln \frac{3}{2}}{f(\frac{1}{2})} = \frac{2(\ln 2 + s \ln \frac{3}{2})}{m \tilde{H}_{2/3}} (1 + o(1)),$$

the second form by (1) (a remark-level asymptotic reading, not a certificate). The sign of the actual deviation is not fixed a priori: per level the shortcut inflates both moments, and the aggregate can move either way. The bias is $O(1/m)$ independently of η — an additive accuracy floor of exactly the kind Theorem 5.1 excludes — and equal-surprisal blocks are the generic structure wherever the exact engine applies (i.i.d. advice over k symbols has $\text{poly}(m)$ levels for k^m keys), which is why the estimator (6) carries $\tilde{\mathcal{A}}_\rho$ throughout.

E.2 Merge cost and floating-point stability

PROOF OF LEMMA 4.3. (i). As in the proof of Theorem 4.1: supports of disjoint blocks add, so the outputs at any one tree level occupy at most $R/\eta + 2m$ grid points in total, and sizing each FFT to its output support makes a level cost $O((R/\eta + m) \log(R/\eta + m))$; over $\lceil \log_2 m \rceil$ levels this is $\tilde{O}(R/\eta + m)$. A depth-first traversal keeps one pending sibling buffer per level, so space is dominated by the root vector, $\Theta(R/\eta + m)$, and the cumulative sum is linear.

(ii). Write λ_i for the per-coordinate log-MGF of the chain being merged and $\Lambda_B = \sum_{i \in B} \lambda_i$ for a block B of $g = |B|$ coordinates; by Lemma 4.2 the exact law at node B is the tilted partial convolution $\mu_B^{(\phi)}$, with analytically known mean $M_B = \Lambda'_B(\phi)$.

Per-coordinate curvature is $O(1)$, uniformly in the range. Set $\lambda_0 := \gamma_0/2$ and let $|\xi| \leq \lambda_0$. In both chains the tilted atom weights decay exponentially in the binned surprisal: writing $u_a := \tilde{s}_i(a) - \tilde{s}_i(a_{\min}) \geq 0$ with $a_{\min}$ the minimal-surprisal symbol, the count chain has tilted atom weights $\omega_a \propto e^{(\phi+\xi)\tilde{s}_i(a)}$ and the mass chain $\omega_a = p_{i,a} e^{(\phi+\xi)\tilde{s}_i(a)}$ with $e^{-\tilde{s}_i(a)} \leq p_{i,a} \leq e^\eta e^{-\tilde{s}_i(a)}$, so in either case

$$\omega_a \leq e^\eta e^{-\gamma u_a} \omega_{a_{\min}}, \quad \gamma := \begin{cases} -(\phi + \xi) & \text{(count chain),} \\ 1 - (\phi + \xi) & \text{(mass chain).} \end{cases}$$

Centring at $\tilde{s}_i(a_{\min})$,

$$\begin{aligned} \lambda_i''(\phi + \xi) &= \text{Var}_w(\tilde{s}_i) \leq \sum_a u_a^2 e^\eta e^{-\gamma u_a} \\ &\leq e^\eta b_i \sup_{u \geq 0} u^2 e^{-\gamma u} = \frac{4b_i e^{\eta-2}}{\gamma^2}, \end{aligned}$$

and $\gamma \geq \gamma_0 - \lambda_0 = \gamma_0/2$ on the whole perturbation window, so $\lambda_i'' \leq \bar{v}_i := 16b_i e^{\eta-2}/\gamma_0^2 = O(b_i/\gamma_0^2)$ — per-coordinate; no uniform alphabet bound enters. No bound on the surprisal range enters either: a symbol of surprisal Σ carries tilted weight $e^{-\gamma\Sigma}$. This negative effective tilt is what makes the bound range-independent, and it is exactly what part (i), which works untilted, lacks. The working tilts are the saddle-centring ones: count chain at $\phi = -1/(1+\rho) \in [-\frac{2}{3}, -\frac{1}{2}]$ and mass chain at $\phi = \rho/(1+\rho) \in [\frac{1}{3}, \frac{1}{2}]$ for $\rho \in [\frac{1}{2}, 1]$ (the mass chain at ϕ is the count chain at $\phi - 1$ up to per-bin factors $e^{O(\eta)}$, by (3)), so $\gamma \geq \frac{1}{2}$ throughout, and we may take $\gamma_0 = \frac{1}{2}$.

Block concentration. For $|\xi| \leq \lambda_0$ we have the exact identity

$$\begin{aligned} \mathbb{E}_{\mu_B^{(\phi)}} [e^{\xi(V-M_B)}] &= e^{\Lambda_B(\phi+\xi) - \Lambda_B(\phi) - \xi \Lambda'_B(\phi)} \\ &= e^{\xi^2 \Lambda''_B(\phi+\xi)/2} \leq e^{\xi^2 V_B/2}, \quad V_B := \sum_{i \in B} \bar{v}_i, \end{aligned}$$

for some ξ between 0 and ξ (Taylor–Lagrange on the smooth convex Λ_B), with $V_B = O(\sum_{i \in B} b_i) = O(B_B)$. Markov and optimizing $\xi \in$

$[0, \lambda_0]$ give the two-regime bound

$$\mathbb{P}_{\mu_B^{(\phi)}}(|V - M_B| \geq u) \leq 2 \exp\left(-\min\left(\frac{u^2}{2V_B}, \frac{\lambda_0 u}{2}\right)\right),$$

so with $L_\epsilon := \ln(2/\epsilon_{\text{tr}})$ the window $W_B := [M_B \pm u_B]$, $u_B := \sqrt{2V_B L_\epsilon} + 2L_\epsilon/\lambda_0$, carries all but ϵ_{tr} of $\mu_B^{(\phi)}$; it spans $O((\sqrt{V_B L_\epsilon} + L_\epsilon)/\eta) = \tilde{O}(\sqrt{B_B}/\eta)$ grid points, which is the window claimed in part (ii).

Truncated merge and error. Evaluate the tree depth-first: at each node convolve the two truncated child vectors by an FFT sized to the sum of their supports, then zero every entry outside W_B , taking $\epsilon_{\text{tr}} := (m/\eta)^{-A}/(2m)$ with A a fixed constant exceeding the polynomial degrees in Lemma 4.4. By induction the working vector obeys $0 \leq \hat{\mu}_B \leq \mu_B^{(\phi)}$ entrywise (convolution of dominated nonnegative vectors is dominated, and truncation preserves this), so each truncation discards at most the *true* out-of-window mass $\leq \epsilon_{\text{tr}}$; and since $\|a * b - \hat{a} * \hat{b}\|_1 \leq \|a - \hat{a}\|_1 + \|b - \hat{b}\|_1$ for sub-probability vectors, the ℓ_1 deficits add over the $2m - 1$ nodes: $\|\hat{\mu}_{\text{root}} - \mu^{(\phi)}\|_1 \leq (2m - 1)\epsilon_{\text{tr}} \leq (m/\eta)^{-A}$. This is an additive one-sided (mass-deficit) perturbation, of size $\leq (m/\eta)^{-A}$ in every norm, and it is absorbed by Lemma 4.4 upon enlarging A : the root window contains the $K\sigma$ assembly window of Lemma 4.4 once A is large enough; pointwise, the contributing saddle entries have size $\text{poly}^{-1} \Theta(\eta/\sqrt{m})$, so the deficit is a further poly^{-1} relative input error; and in the cumulative sums of the count chain the analytic reweighting $e^{|\phi|w}$ of (5) is increasing in w , so the discarded tail below a contributing level t contributes at most $\epsilon_{\text{tr}} e^{|\phi|t}$ against a retained sum $\geq e^{|\phi|t} \text{poly}^{-1} \Theta(\eta/\sqrt{m})$ — again relative poly^{-1} . (The mass chain enters the assembly only pointwise.)

The d -dimensional merge. The finite-rank engine of Section 4.1 mirrors this construction with $\mathbb{Z}^d$ in place of the grid: per-coordinate supports add along each axis, so the nodes of any one round jointly hold $O(m^d)$ cells, each d -dimensional FFT is sized to the actual support box of its output, and the merge costs $\tilde{O}(m^d)$ in total; the truncation and domination arguments above apply per axis unchanged.

Cost. With $L_\epsilon = O(\ln(m/\eta))$, node B holds $n_B = O((\sqrt{V_B L_\epsilon} + L_\epsilon)/\eta)$ points; over the 2^j nodes of level j , Cauchy–Schwarz gives $\sum_B \sqrt{V_B} \leq \sqrt{2^j \sum_B V_B} = O(\sqrt{2^j B_{\text{in}}})$, so the level costs $\tilde{O}((\sqrt{2^j B_{\text{in}}} + 2^j L_\epsilon)/\eta)$, geometric in j and dominated by the leaf level: $\tilde{O}((\sqrt{m B_{\text{in}}} + m)/\eta)$ in total — $\tilde{O}(m/\eta)$ for bounded alphabets — with the alphabet sizes entering only through B_{in} . The depth-first traversal holds one buffer per level, for $\tilde{O}(\sqrt{B_{\text{in}}}/\eta)$ space in total, and the final cumulative sum is linear in the root support $\tilde{O}(\sqrt{B_{\text{in}}}/\eta)$. $\square$

PROOF OF LEMMA 4.4. We prove the floating-point bound and the absence of underflow, arguing for Algorithm 2 (binned, two chains); Algorithm 1 is the special case with the single-chain collapse (3), and it is strictly better conditioned. Throughout, $L = \tilde{O}(\sqrt{B_{\text{in}}}/\eta)$ is the truncated transform length (Lemma 4.3 (ii)), and poly abbreviates a fixed polynomial in $m, 1/\eta$.

Per-FFT error. For a radix-2 FFT of length L whose weights carry normwise error $\mu = O(u_m)$ (hypothesis (24.2) of [21]; standard libraries satisfy this), Higham’s bound [21, Thm. 24.2] gives normwise relative error $O(u_m \log L)$, and the same for the inverse transform; a convolution $a * b = \text{IFFT}(\text{FFT}(a) \odot \text{FFT}(b))$ inherits $O(u_m \log L)$.

Conditioning of convolution. For probability vectors $a, b \geq 0$ with $\|a\|_1 = \|b\|_1 = 1$, Young’s inequality gives $\|a * b\|_2 \leq \|b\|_2$ and $\|a * b\|_1 = 1$: convolution is non-amplifying. When input errors are propagated, the FFT bound is normwise in ℓ_2 ; the conversion $\|\delta a\|_1 \leq \sqrt{L}\|\delta a\|_2$ introduces an $O(\sqrt{L})$ factor per step, absorbed into poly.

Merge-tree composition. The m leaves combine by $m - 1$ convolutions in a balanced tree of depth $\lceil \log_2 m \rceil$. Relative errors do *not* merely compose along one root-to-leaf path: at every parent the errors of both children add — for $c = a * b$ with $\tilde{a} = a(1 + \alpha)$, $\tilde{b} = b(1 + \beta)$, entrywise $|\alpha|, |\beta| \leq E$, one gets $E_{\text{parent}} \leq E_a + E_b + \varepsilon_{\text{node}}$, and every quantity in the tree is a non-negative count or mass, so no cancellation can be appealed to — and the root accumulates a contribution from each of the $m - 1$ internal nodes. A node’s relative contribution at the root is scaled by $\|v_{\text{node}}\|_2 / \|v_{\text{root}}\|_2$, which is largest at the leaves ($\approx \sqrt{L}$); summing over the levels gives a geometric series dominated by the leaf level, so

$$\|\text{fl}(v^{(\phi)}) - v^{(\phi)}\|_2 \leq \varepsilon_{\text{tree}} \|v^{(\phi)}\|_2,$$

with $\varepsilon_{\text{tree}} = O(u_m m \sqrt{L} \log L) = O(u_m \text{poly})$. The node-count factor m is essentially tight: at fixed root support the measured growth exponent in m is 0.96, while the $\sqrt{L}$ amplification is the conservative part. Measured end to end, the root error is 3.3×10^{-15} at $m = 512$ in double precision — negligible, but the budget is $m/\log m$ ($\approx 32 \times$ at $m = 256$) larger than a depth-only count would suggest, and it is this corrected value that enters poly. The same leaf-dominated accumulation, over the same $\Theta(m)$ internal nodes, covers the d -dimensional merge of the finite-rank engine (Section 4.1).

Normwise $\rightarrow$ entrywise on the saddle entries. The tilted law $v^{(\phi)}$ is concentrated: its peak is $\|v^{(\phi)}\|_\infty = \Theta(\eta/\sqrt{m})$, so that $\|v^{(\phi)}\|_2 \leq \Theta(\eta^{1/2} m^{-1/4})$. Entries feeding the assembly (within $K\sigma$ of the center, tolerance poly^{-1}) have size $g \geq \text{poly}^{-1} \|v^{(\phi)}\|_\infty$, so their relative error is $\varepsilon_{\text{tree}} \|v^{(\phi)}\|_2 / g = O(u_m \text{poly})$. This is the one nonstandard step: it works precisely because the saddle entries are within a poly factor of the ℓ_2 norm.

Log-domain assembly. By (5) we assemble $\ln \tilde{F}(t) = \Lambda_U(\phi) + \text{LSE}_{w \leq t}[-\phi w + \ln v^{(\phi)}(w)]$. The inputs carry: $-\phi w$, absolute error $O(mu_m)$; $\ln v^{(\phi)}(w)$, absolute error $O(u_m \text{poly})$; $\Lambda_U(\phi)$, absolute error $O(mu_m)$. Since LSE is 1-Lipschitz in ℓ_∞ , $\ln \tilde{F}(t)$ has absolute error $O(u_m \text{poly})$, and likewise $\ln \tilde{\pi}(t)$. One further 1-Lipschitz LSE gives $f(\rho)$ with absolute error $O(u_m \text{poly})$.

Final ratio. Write $\hat{f}_\eta(\rho) = f_\eta(\rho) + e_\rho$ with $|e_\rho| = O(u_m \text{poly})$. Since $s_\eta = f_\eta(1)/f_\eta(\frac{1}{2})$, $\hat{s} - s_\eta = \frac{e_{1-s_\eta e_{1/2}}}{f_\eta(\frac{1}{2}) + e_{1/2}}$. Provided $|e_{1/2}| \leq$

$f_\eta(\frac{1}{2})/2$, this gives $|\hat{s} - s_\eta| = O\left(\frac{u_m \text{poly}(1+s_\eta)}{f_\eta(\frac{1}{2})}\right)$. For families with $f_\eta(\frac{1}{2}) = \Theta(m)$ and $s_\eta = O(1)$, this simplifies to $O(u_m \text{poly}/m)$.

No underflow. Every materialized float is either (i) an entry $v^{(\phi)}(w) \in [0, 1]$ of a tilted probability vector, whose smallest contributing value is $\geq \text{poly}^{-1} \|v^{(\phi)}\|_\infty$, i.e. of size $\Theta((\eta/\sqrt{m}) \text{poly}^{-1})$, far above the IEEE double underflow threshold for any practical $m, 1/\eta$; or (ii) a log-domain quantity of magnitude $O(m)$. The rare-event magnitudes $e^{-\Theta(m)}$ enter only through their logarithms, via the analytic factor $-\phi w + \Lambda_U(\phi)$, and are never exponentiated. $\square$

The explicit chain, and the floor that binds. Every step above has explicit constants available: Higham’s per-FFT bound, the corrected leaf-dominated tree composition, the $\ell_2 \rightarrow$ entrywise conversion via the concentration estimate $\|v^{(\phi)}\|_\infty = \Theta(\eta/\sqrt{m})$, the 1-Lipschitz log-domain assembly (including the $\text{expm1}/\log 1p$ evaluation of the level averages, Appendix E.1), and the final division by $f_\eta(\frac{1}{2}) = \Theta(m)$. Working from the measured root error $\approx 10^{-14}$: the induced error in s is $\approx (1+s) \cdot 10^{-14}/f(\frac{1}{2}) \approx 2 \times 10^{-16}$ at $m = 256$; it grows as η shrinks (at most like $\sqrt{L} \propto \eta^{-1/2}$) and crosses the binning error $C(s)\eta$ at $\eta_{\min} \sim 10^{-11}$, i.e. a *floating-point floor* $\delta_{\min} \sim 10^{-10}$ in IEEE double. For heterogeneous advice this — not the atom spacing ($\sim 10^{-120}$ at $m = 256$) and not the local CLT’s $e^{-\Theta(m)}$ — is the floor that operates; for low-rank advice the atom spacing ($\Theta(m^{1-k})$) binds first and floating point is second (Remark 6.10). Both sit far below any target of practical interest, so the honest reading of “computable in IEEE double precision” is: the stability analysis supports targets down to $\sim 10^{-10}$ in the tested regime; rigorous end-to-end certification requires the interval route at higher working precision, with bit complexity scaling accordingly. One regime deserves a check when implementing: high-confidence advice drives $f(\frac{1}{2}) \rightarrow 0$ and amplifies the induced error $(1+s)\varepsilon/f(\frac{1}{2})$ — exactly where the tilting works hardest; the certificate reports the amplification automatically, since it divides by the computed $f_\eta(\frac{1}{2})$.